A Data-Driven, Component-Level Projection Framework for 21st-Century Sea-Level Rise

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Abstract

Every centimeter of sea-level rise exposes about 1.6 million more people to annual coastal flooding, so planning depends on forecasts whose skill can be tested against observations. We build SPECTRUM (Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models), which represents each contributor to global mean sea level (GMSL) with a parsimonious physical model. Thermal expansion, mountain glaciers, the Antarctic Peninsula, and Greenland ice discharge are calibrated by Bayesian inference against independent multi-decade records. Greenland surface mass balance is emulated, and East Antarctic surface mass balance and terrestrial water storage are adopted from published work. Observed GMSL is withheld from every calibration and used only as a test. Forced with observed global mean surface temperature, the summed components close 94–103% of the GMSL rate from satellite altimetry since 1993. Their 1950–2000 rate of rise lies within the range of observational records, and they reproduce how the rate of rise has changed with warming, including the shift from ocean warming to land ice and water as the larger contributor. The 1950–2025 record spans 1.1–1.2 °C of warming, comparable to warming left this century in the 2.7 °C scenario, which supports forecasts of the calibrated components through 2100. The framework cannot constrain West Antarctic mass loss, which we represent with slow and fast storylines. With slow loss assigned 90% probability, median GMSL rise over 2000–2100 is 0.46, 0.60, and 0.78 m under 1.8, 2.7, and 3.6 °C of late-century warming. Although fast loss is assigned only 10% probability, West Antarctica accounts for 73–94% of forecast variance.

1 Introduction

Sea-level rise will continue for centuries even if warming stops, so the question for coastal communities is not whether to adapt but how fast (*Fox-Kemper et al.*, 2021). Each centimeter of rise exposes roughly 1.6 million more people to annual coastal flooding (our fit to the data of *Kulp and Strauss* 2019). Without further adaptation, global flood damages could reach US$14 trillion per year at 0.86 m of rise (*Jevrejeva et al.*, 2018). Coastal planners use widely varying projections (*Hirschfeld et al.*, 2023). The Sixth Assessment Report of the Intergovernmental Panel on Climate Change (AR6) projects median rise of 0.44 and 0.56 m by 2100, relative to 1995–2014, for low- and intermediate-emissions scenarios (SSP1-2.6 and SSP2-4.5) (*Fox-Kemper et al.*, 2021). AR6 also attributes deep uncertainty to ice-sheet

processes that models do not represent well (*Fox-Kemper et al.*, 2021). High-end estimates published since AR6 span 0.6–2.0 m of rise by 2100 (*Garner et al.*, 2026).

Whatever the method behind these projections, continued warming ensures that they will remain out of sample with respect to the observational record. The central difficulty is therefore not building a model but establishing when its projections can be trusted. A model earns that trust when it is calibrated with observations and tested against observations withheld from calibration. Projections can also be tested against observations made after they were published. The mid-range projection of the IPCC Second Assessment Report matched the global sea-level rise of the following three decades to within about 10 mm, while its projections of individual contributors were more variable (*Törnqvist et al.*, 2025). A test against withheld data is most informative when the withheld record spans a change in forcing comparable to the change being forecast. Such a test does not remove the out-of-sample problem, but it estimates how far a model can be carried beyond the conditions it was calibrated on. A model must also report its uncertainties in a form that separates their sources (*Aschwanden et al.*, 2021). A recent community synthesis makes this case, calling for projections narrowed by observations and for uncertainty characterized by its structure and not only by its magnitude (*Felikson et al.*, 2025).

Sea-level projection models span a wide range of complexity, and they meet this standard to different degrees, but none fully. At the most detailed end are process-based ice-sheet models, which solve the equations of ice flow and deformation and can test competing hypotheses about the mechanics of ice loss (*Schoof*, 2007; *Morlighem et al.*, 2024). They are needed to project ice loss under conditions outside any observational record. Reproducing observed mass change is a necessary, though not sufficient, condition for trusting their projections (*Aschwanden et al.*, 2021), and this condition is hard to meet. One obstacle is that their parameters are inferred by solving inverse problems in which the data constrain only a few combinations of parameters (*Petra et al.*, 2014; *Isaac et al.*, 2015). As of 2020, no ice-sheet model run forward in time had reproduced the sixfold increase in ice-sheet mass imbalance observed since the early 1990s (*Siegert et al.*, 2020). Most of the recent Antarctic projections grow the ice sheet over the first few years of the simulations, while observations over the same years show it losing mass (*Fricker et al.*, 2025; *Seroussi et al.*, 2024). Projections also depend on how each model is initialized and calibrated (*Seroussi et al.*, 2020; *Goldberg et al.*, 2026).

Models run to equilibrium with past climate before the projection begins change little on their own but miss the observed present-day mass loss. Models initialized to match the present geometry and flow may not reproduce the observed rate of mass change (*Pattyn and Morlighem*, 2020; *van den Akker et al.*, 2025). In the Antarctic ensemble of the Ice Sheet Model Intercomparison Project (ISMIP6), the choice of ice-sheet model accounts for 52% of the uncertainty in projected dynamic mass loss by 2100, compared with 13% for the choice of climate model (*Seroussi et al.*, 2023), and the projected Antarctic contribution to 2100 shows no clear response to the emissions scenario (*Edwards et al.*, 2021). These models also lack a validated representation of iceberg calving and fracture, and many hold the calving front fixed (*Pattyn and Morlighem*, 2020; *Bassis et al.*, 2024; *Benn et al.*, 2026). AR6 complements these models with structured expert judgment in its low-confidence projections (*Bamber et al.*, 2019; *Fox-Kemper et al.*, 2021).

At the other extreme are approaches anchored directly in the observed record of global

mean sea level (GMSL). The simplest extrapolates the observed rate and acceleration of GMSL rise forward in time (*Nerem et al.*, 2022). Semi-empirical models instead fit the rate of GMSL rise to global mean surface temperature (GMST) (*Rahmstorf*, 2007; *Vermeer and Rahmstorf*, 2009; *Rahmstorf et al.*, 2012; *Kopp et al.*, 2016). Both are transparent and fit the quantity being predicted directly. However, they give no insight into which processes drive the rise. The fitted relationship also need not hold when contributors with different response times change their relative importance (*Church et al.*, 2013).

Between these extremes sit component-level models, which constrain each physical contributor separately, trading spatial detail for parsimony and transparency (*Perrette et al.*, 2013; *Kopp et al.*, 2014; *Mengel et al.*, 2016; *Nauels et al.*, 2017; *Wong et al.*, 2017; *Kopp et al.*, 2023). Component-level frameworks differ in how each contributor is constrained and in whether the observed GMSL record is used in calibration or kept for testing. When every calibrated component is fit to its own independent record and GMSL is withheld throughout, as in *Mengel et al.* (2016), the sum of the calibrated components can be tested against observed GMSL.

No single class of model can be at once the most detailed in physical processes, the most transparent, and the most tightly constrained by data. Ordered by complexity, models can instead form a hierarchy in which each level is calibrated and tested against data before the next level of complexity is trusted. Ordering models from idealized to comprehensive is long established in climate science, where it has helped close the gap between simulation and understanding (*Held*, 2005; *Jeevanjee et al.*, 2017; *Maher et al.*, 2019). Such a hierarchy does not require any one model to be correct. It requires only that its models rest on assumptions different enough that their agreement or disagreement is informative. Where structurally different models agree, a result rests on more than one set of assumptions (*Lipscomb et al.*, 2025). Where they disagree, the divergence shows what the data leave unconstrained and which observations would constrain it.

Simpler levels also bring practical advantages. Across 97 published comparisons, simple forecasting methods were as accurate as or more accurate than complex ones in 84% of cases (*Green and Armstrong*, 2015). In a competition on 100,000 time series, about two-thirds of the submitted methods did not beat a simple combination of exponential-smoothing forecasts (*Makridakis et al.*, 2020). The assumptions of simple models are also easier to state and check, which helps those who use the results find errors and omissions (*Green and Armstrong*, 2015). A hierarchy built this way complements process-based ice-sheet models and aids their development rather than substituting for them.

Here we construct a data-driven, component-level forecast of 21st-century sea-level rise. We call the framework SPECTRUM, for Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models. Each contributor that the observational record can constrain is represented by a parsimonious physical model and calibrated by Bayesian inference against its own independent record. Among these, Greenland ice discharge is modeled as a lagged response to the temperature of the surrounding ocean, calibrated with observed discharge. Observed GMSL is withheld from every calibration. The summed components can therefore be tested against observations from 1950 to 2025, a period spanning 1.1–1.2 °C of warming, comparable to the warming that remains this century under intermediate emissions. We place SPECTRUM in a hierarchy and compare its forecast with extrapolation of the observed trend, a semi-empirical model, and the AR6 assessment. Where the record

cannot constrain a contributor, as for the West Antarctic Ice Sheet (WAIS), we say so and represent the range of outcomes with storylines rather than a calibrated model, keeping this source of uncertainty separate from the others. This separation lets us ask which part of 21st-century sea-level rise can be reduced by cutting emissions. It also shows which part of the uncertainty can be narrowed only by observations and models that constrain how fast West Antarctica will lose ice.

2 A data-driven, component-level projection framework

SPECTRUM represents each contributor to GMSL with its own model, and the way each model is constrained depends on the data available (Fig. 1). Thermal expansion, mountain glaciers, the Antarctic Peninsula, and Greenland ice discharge are calibrated by Bayesian inference against independent multi-decade observational records, listed in Table S1 of the Supplementary Material. Greenland surface mass balance is emulated from regional climate model simulations. East Antarctic surface mass balance and terrestrial water storage are adopted from published sensitivities and assessments, and West Antarctica is represented by slow and fast storylines. Each temperature-sensitive component is forced by GMST, directly or through a regional temperature derived from it, and its temperature dependence, calibrated or adopted, defines its forecast. Observed GMSL is withheld from the calibrations and used to test the sum of the components.

Earlier component-level frameworks constrain their contributors in different ways. Several draw their components from process models, assessments, or expert judgment. *Kopp et al.* (2014) build the distribution of each contributor from process-model ensembles, the AR5 assessment, and expert elicitation. The MAGICC sea level model (*Nauels et al.*, 2017) emulates process-based projections, with each component calibrated with output from climate, glacier, or ice-sheet models. FACTS (*Kopp et al.*, 2023) draws its land-ice contributions from emulators of model intercomparison projects or from structured expert judgment. Others use observed GMSL in calibration. BRICK (*Wong et al.*, 2017) calibrates its component models against observations and then conditions the assembled ensemble on observed GMSL. In the high ice-sheet case of *Perrette et al.* (2013), the ice-sheet contribution is the residual of a semi-empirical model calibrated with observed GMSL. The design closest to SPECTRUM is that of *Mengel et al.* (2016), who calibrate a semi-empirical relation for each contributor against its own observations, constrained by estimates of long-term commitment, and compare the sum with total sea-level reconstructions not used in the calibration. SPECTRUM follows that design and differs from it in three ways. Its parameters are estimated by Bayesian inference, which yields their joint posterior distribution, rather than by least squares. Greenland ice discharge is forced by the temperature of the ocean around Greenland, whereas *Mengel et al.* (2016) drive it with GMST. Finally, *Mengel et al.* (2016) note that their method cannot represent processes that proceed independently of the forcing, such as a collapse of the Amundsen Sea sector of West Antarctica. SPECTRUM represents the range of West Antarctic outcomes with storylines instead.

Unless otherwise noted, all sea-level values in the main text are global-mean 21st-century contributions referenced to the year 2000. Throughout, we label each emissions scenario by the AR6 central estimate of its global mean surface temperature over 2081–2100 relative to

the 1850–1900 (pre-industrial) mean (*Lee et al.*, 2021, Table 4.5): 1.4 °C (SSP1-1.9), 1.8 °C (SSP1-2.6), 2.7 °C (SSP2-4.5), 3.6 °C (SSP3-7.0), and 4.4 °C (SSP5-8.5). The central paths of the 1.8, 2.7, and 3.6 °C scenarios reach 1.7, 2.7, and 3.9 °C in 2099, the last year of the AR6 projections. We hold each path at its 2099 value thereafter.

2.1 Component models, calibration, and validation

2.1.1 Greenland Ice Sheet

The Greenland Ice Sheet (GrIS) is among the largest land-ice contributors to sea-level rise and has lost mass every year since 1998 (*Mouginot et al.*, 2019). That loss is the imbalance between solid-ice discharge and surface mass balance (SMB), where SMB is the difference between mass gained by snow accumulation and mass lost to meltwater runoff and sublimation. GrIS discharge is sensitive to ocean temperature and surface melt runoff (*Straneo et al.*, 2013; *Slater et al.*, 2020; *Wood et al.*, 2021). Warm subsurface waters thin and undercut the calving fronts of marine-terminating glaciers, driving increased calving and net terminus retreat. This effect is amplified where surface runoff drives mixing and buoyant upwelling along the calving face (*Slater and Straneo*, 2022). Surface melt also lubricates glacier beds, but that effect is generally short-lived and localized as the subglacial hydrological system evolves to accommodate changes in meltwater flux (*Schoof*, 2010; *Tedstone et al.*, 2013; *Ultee et al.*, 2022). We therefore model discharge as a lagged response to ocean temperature.

We approximate the GrIS-wide discharge as

$$D = \gamma\, T_{\mathrm{ocean}}(t - \delta) + r_0, \tag{1}$$

where D is the GrIS discharge in sea-level equivalent (SLE) units, γ is an effective discharge sensitivity to ocean temperature, δ is the lag between warming of the ocean around Greenland and the response of discharge, r_0 is the background discharge rate at the baseline ocean temperature, and T_{ocean} is the ocean temperature anomaly in the waters around Greenland relative to the 1995–2005 mean. SLE is the rise in global mean sea level that would result if a given ice mass were added to the ocean, with 1 mm SLE = 362.5 Gt throughout this study. Because D is the gross discharge flux rather than an anomaly, it becomes a contribution to GMSL only when combined with SMB. We emulate SMB separately from a regional climate model (Section 2.2.1).

We calibrate against total GrIS discharge, summed over the 219 glaciers classed as tidewater in the *Mouginot et al.* (2019) dataset, of which 107 (85% of discharge) are measured directly. Summing gives up spatial detail but reduces noise relative to the signal. Our goal is to capture ice-sheet-scale discharge rather than that of individual basins and glaciers. The calibration is carried out on the time-integrated form of Eq. 1, which gives the cumulative discharge,

$$H_{\mathrm{D}}(t) = \int_{t_0}^{t} D(t')\,\mathrm{d}t', \tag{2}$$

with γ and r_0 held constant in time. Fitting the cumulative record reduces the influence of single-year noise and directly constrains the total mass lost, the quantity of interest in this study. Its errors are correlated in time, which we account for below.

The calibration uses the ice discharge time series of *Mouginot et al.* (2019), which spans the 47 years from 1972 to 2018, and ocean temperatures from EN4 (v4.2.2) (*Good et al.*, 2013). We average EN4 over 200–500 m depth, the range most relevant to the grounding lines of Greenland's tidewater glaciers (*Slater et al.*, 2020), and over the seas surrounding Greenland south of 80°N (58–80°N, 75–5°W). This geographic limit is set by the sparse subsurface sampling farther north. EN4 begins in 1970, so the delayed forcing is undefined in the first years of the discharge record, and every candidate lag is fit to the same 41 years, 1978 to 2018, those covered by the longest lag.

Because the data are resolved annually, the lag δ takes integer-year values. We fit (γ, r_0) by weighted least squares separately for each candidate lag from 4 to 8 yr, using the demeaned cumulative data (Fig. 2). Each fit carries a penalty on the mismatch between the modeled discharge rate at the end of the time series and the rate observed over the final decade, which keeps the calibration consistent with the present-day discharge rate. The posterior is then formed as a Gaussian mixture of the fits for each δ, with weights from the Bayesian information criterion. Every candidate lag has the same number of fitted parameters, so these weights reflect only how well each lag fits the data. This procedure places 98.8% of the weight on $\delta = 5$ yr among the candidates and gives posterior medians $\gamma = 0.367$ [0.313, 0.421] $\mathrm{mm\,SLE\,yr^{-1}\,{}^{\circ}C^{-1}}$ and $r_0 = 1.386$ [1.369, 1.403] $\mathrm{mm\,SLE\,yr^{-1}}$ (Fig. S1). The weights are computed before the corrections for correlated errors and excess scatter described next, so the 98.8% is an upper bound. The residuals scatter more widely than the reported observational errors allow, with reduced chi-square $\chi^2/\mathrm{dof} = 2.0$ on 38 degrees of freedom, where χ^2 is the error-weighted sum of squared residuals. Because the calibration target is a cumulative record, its errors are also correlated in time. We therefore report parameter intervals from a covariance that accounts for both effects (Supplementary Material).

The fitted $\delta = 5$ yr is an effective lag for spatially averaged forcing, and we therefore do not attempt to attribute it to a single process. It lies within the range of timescales spanned by the ocean-to-ice pathway. These timescales include the exchange of water between the continental shelf and the fjords, the response of glacier fronts to undercutting, and the inland spread of retreat to the flux gates, the fixed cross-sections near each glacier front where discharge is measured (*Straneo and Heimbach*, 2013; *Wood et al.*, 2021; *Robel et al.*, 2018). Because T_{ocean} is averaged over Greenland's peripheral waters, δ also absorbs the time required for a warming anomaly to become representative of conditions at outlet glaciers distributed around the ice sheet. Eq. 1 is linear and reversible, so it does not represent retreat that persists after ocean forcing declines, nor the control of fjord and bed topography on retreat. Glaciers calving on shallow ridges or standing in cold, shallow water have retreated little, while glaciers in deep fjords reached by warm Atlantic Water have retreated substantially (*Wood et al.*, 2021).

We validate the model against two records withheld from calibration, the discharge time series of *Mankoff et al.* (2021) spanning 1986–2023 and the time series from the Ice Sheet Mass Balance Intercomparison Exercise (IMBIE-3) spanning 1972–2023 (*Otosaka et al.*, 2026). The validation with *Mankoff et al.* (2021) is informative but weak, because the *Mankoff et al.* (2021) time series shares its underlying raw data with the *Mouginot et al.* (2019) calibration and differs from it only in processing. IMBIE-3 combines three kinds of estimates. Input-output estimates subtract discharge from SMB, and those in IMBIE-3 include *Mouginot et al.* (2019) and *Mankoff et al.* (2021). Altimetry estimates, from satellite measurements

of ice-surface height since 1992, and gravimetry estimates, from satellite measurements of Earth's gravity field since 2002, are independent of the calibration. Before 1992, IMBIE-3 rests on input-output estimates alone. Its discharge series is total mass balance minus the mean SMB of three regional climate models, one of which is RACMO, the model that also supplies the *Mouginot et al.* (2019) SMB. The independent part of the IMBIE-3 test is therefore the total mass balance after 1992. Our model falls within the uncertainties of the validation data for discharge and total mass balance (Fig. 2). Because the SMB part of the model is fitted to the *Mouginot et al.* (2019) SMB (Section 2.2.1), the total mass balance mainly tests the discharge model. The SMB emulator alone is tested against observed SMB in the Supplementary Material. The validation results suggest that our simple model, with two fitted parameters and a selected lag, captures cumulative GrIS discharge during the observational record and its sensitivity to ocean temperatures in Greenland's peripheral waters.

To our knowledge, no published study reports a Greenland-wide sensitivity of discharge rate to ocean temperature, so we derive a simple dimensional scaling to contextualize the inferred value $\gamma = 0.367$ mm SLE yr^{-1} °C^{-1}. For a single outlet glacier i, the submarine melt rate scales approximately linearly with ocean thermal forcing, the ocean temperature above the in situ freezing point. We treat a change in T_{ocean} as an equal change in thermal forcing, so that

$$\Delta \dot{m}_i = \kappa_i \, \Delta T_{\mathrm{ocean}}, \tag{3}$$

where κ_i is the melt sensitivity to thermal forcing averaged over the submerged calving face. Plume parameterizations calibrated with Greenland observations (*Xu et al.*, 2013; *Rignot et al.*, 2016; *Slater and Straneo*, 2022) give a maximum melt sensitivity of 90–260 m yr^{-1} °C^{-1} near the grounding line, where the glacier loses contact with its bed. The face-averaged rate is two to three times smaller (*Wood et al.*, 2021). Dividing by 2.5, we take $\kappa_i = 36$–104 m yr^{-1} °C^{-1}, with a median of 60 m yr^{-1} °C^{-1}. Submarine melt thins and undercuts the terminus, which triggers calving and retreat of the grounding line. The resulting increase in ice flux is larger than the direct submarine melt flux by a factor μ. Neither flux is measured ice-sheet wide, so we take $\mu = 3$–10 as an assumed range. For comparison, submarine melt beneath floating termini removes 5–85% of grounding-line discharge (*Enderlin and Howat*, 2013), and ice sheet projections that exclude ocean-induced undercutting may underestimate mass loss by at least a factor of 2 (*Wood et al.*, 2021). The flux perturbation from a single glacier with submerged calving-front area A_{f} is then

$$\Delta \dot{Q}_i = \mu_i \, \rho_{\mathrm{ice}} \, A_{\mathrm{f},i} \, \kappa_i \, \Delta T_{\mathrm{ocean}}, \tag{4}$$

where $\rho_{\mathrm{ice}} = 900$ kg m^{-3} is the density of ice. Summing over Greenland's $N = 219$ marine-terminating outlets (*Mouginot et al.*, 2019), we approximate the scale of γ as

$$\widetilde{\gamma} \sim \frac{1}{A_{\mathrm{ocean}} \, \rho_{\mathrm{fw}} \, \Delta T_{\mathrm{ocean}}} \sum_{i=1}^{N} \Delta \dot{Q}_i, \tag{5}$$

where $A_{\mathrm{ocean}} = 3.625 \times 10^{14}$ m^2 is the ocean area and $\rho_{\mathrm{fw}} = 1000$ kg m^{-3} is the density of freshwater. Together, these convert added ice mass to SLE. Submerged calving-front areas of 0.9–2.7 km^2 are reported for five West Greenland outlets (*Rignot et al.*, 2016), and we take

this range as representative of all 219 outlets. Evaluating Eq. 5 across the full parameter ranges gives

$$\widetilde{\gamma} = 0.05\text{–}1.5 \text{ mm SLE yr}^{-1} \,^{\circ}\text{C}^{-1}, \tag{6}$$

which contains our calibrated $\gamma = 0.367$ mm SLE yr^{-1} °C^{-1}. We report Eq. 6 only as an order-of-magnitude check on the inferred value, to build intuition. The range spans a factor of 30 because it multiplies three uncertain quantities, μ, A_f, and κ, none of which is measured across the whole ice sheet. The estimate also treats these quantities as uncorrelated across outlets. In fact, the deepest and widest fronts sit in the warmest fjords, which would push the estimate upward. Conversely, two of the five glaciers with measured front areas are among the fastest in Greenland (*Rignot et al.*, 2016), so the adopted areas likely exceed the mean over all 219 outlets, most of which are small. Finally, surface meltwater that drains beneath glaciers strengthens the plumes that melt their fronts, and this runoff rises along with ocean temperature. The calibrated γ is therefore an effective sensitivity that also captures part of the atmospheric influence on discharge. At the ice-sheet scale, atmospheric variability in submarine melting accounts for about 30% of modeled retreat and dynamic sea-level contribution (*Slater and Straneo*, 2022).

2.1.1.1 Surface mass balance

For Greenland SMB, the forecast uses a statistical emulator of the regional climate model MARv3.12, driven by seven of the CMIP6 global climate models downscaled with MAR by *Boberg et al.* (2026). For each global model, we fit the annual SMB contribution as a quadratic in that model's GMST anomaly T relative to 1995–2005, smoothed with an 11-yr centered mean, with AR(1) residuals, $\dot{H}_\mathrm{smb} = a_\mathrm{gris}T^2 + b_\mathrm{gris}T + c_\mathrm{gris}$, pooling its historical and SSP simulations (Supplementary Material, Figs. S6–S8). Each Monte Carlo member draws one global model, with equal weight, and one posterior draw of a_gris and b_gris. Driven by observed GMST, the emulator does not reproduce a slowly varying departure of the observed SMB from its response to warming. Relative to 1995–2005, the *Mouginot et al.* (2019) SMB lies about 79 Gt yr^{-1} above the emulator over 1972–1985 and 106 Gt yr^{-1} below it over 2010–2018 (Supplementary Material). We represent this departure as a drift state $s(t)$ that relaxes toward zero over a timescale τ_s, so that $\dot{H}_\mathrm{smb} = a_\mathrm{gris}T^2 + b_\mathrm{gris}T + c_\mathrm{gris} + s(t) + \varepsilon(t)$, where ε is year-to-year weather. For each member, c_gris and one path of s are drawn from their Kalman-filter posterior given the *Mouginot et al.* (2019) SMB over 1972–2018. The 47-year record does not constrain τ_s. We therefore set it with a lognormal prior whose 90% range is 5 to 25 yr, which allows about two relaxation times within the record, and obtain $\tau_\mathrm{s} = 12$ [6, 26] yr. In 2018 the observed SMB minus the emulator is -27 [-167, 108] Gt yr^{-1}, and this offset relaxes toward zero thereafter. The drift state does not change the emulator's response to warming. We also include the SMB–elevation feedback, which adds $8 \pm 5\%$ (1σ, truncated at zero) to SMB loss in the reference simulation of *Fettweis et al.* (2013). We scale this fraction by each member's cumulative SMB anomaly since 2000. Because MAR is the only regional climate model in the training set, the spread between regional climate models is not sampled (*Glaude et al.*, 2024). When each global model is left out in turn, the other six predict its 2015–2100 SMB contribution within the 90% predictive range for all 13 withheld simulations.

2.1.2 Thermosteric

The ocean is the dominant reservoir for the excess heat accumulated by the climate system (*Cheng et al.*, 2017), and the resulting thermal expansion of seawater, the thermosteric contribution, is currently the largest single contributor to observed sea-level rise (*Frederikse et al.*, 2020). A simple model of thermal expansion must represent both heat exchange between the ocean and the atmosphere and the transport of heat within the ocean. The ocean warms with a delay relative to the atmosphere. Heat enters at the sea surface and is carried downward by turbulent mixing and by the overturning circulation, so the temperature of the interior reflects the history of surface warming. Downward transfer spans a wide range of timescales. The upper few hundred meters exchange heat with the atmosphere over years to decades, whereas water below the thermocline is renewed over centuries. We therefore represent the ocean as two layers, a fast upper layer coupled to the atmosphere and a slow deep layer coupled only to the upper layer, following the structure of *Geoffroy et al.* (2013). This structure reproduces both the lag between surface warming and expansion and the continued expansion that follows a stabilization of surface temperature, because the deep layer continues to draw heat from the upper layer after the upper layer has equilibrated.

Consider first an upper layer of thickness $h_\mathrm{u} = 700$ m warmed by $\Delta\theta(z,t)$. The resulting rise in the steric height of the ocean is

$$H_\mathrm{u} = \int_0^{h_\mathrm{u}} \int_0^{\Delta\theta} \alpha_\mathrm{sw}\, d\theta'\, dz, \tag{7}$$

where $\alpha_\mathrm{sw}(z)$ is the thermal expansion coefficient of seawater, which varies with depth because it depends on temperature and pressure. The temperature anomaly profile is defined as $\Delta\theta = S_\mathrm{u}\varphi$, where $S_\mathrm{u}(t)$ is the amplitude of the anomaly and $\varphi(z)$ is a dimensionless shape factor. Expanding $\alpha_\mathrm{sw} = \alpha_{0,\mathrm{sw}}(z) + \alpha'_\mathrm{sw}\theta'$, with $\alpha'_\mathrm{sw} = \partial\alpha_\mathrm{sw}/\partial\theta$, we rewrite Eq. 7 as

$$H_\mathrm{u} = a_\mathrm{u} S_\mathrm{u}^2 + b_\mathrm{u} S_\mathrm{u}, \tag{8}$$

where $a_\mathrm{u} = \frac{1}{2}\int_0^{h_\mathrm{u}} \alpha'_\mathrm{sw}\varphi^2 dz$ and $b_\mathrm{u} = \int_0^{h_\mathrm{u}} \alpha_{0,\mathrm{sw}}\varphi dz$. The coefficient b_u sets the expansion of the upper layer per degree of warming. In the fit, the quadratic term a_u mainly captures departures from a linear response, such as warming that reaches deeper over time, with a smaller part from the temperature dependence of α_sw. As GMST, $T(t)$, changes, the amplitude evolves as

$$\frac{dS_\mathrm{u}}{dt} = \frac{T - S_\mathrm{u}}{\tau_\mathrm{u}}, \tag{9}$$

where τ_u is the characteristic time over which the thermal state S_u relaxes toward GMST. We scale φ so that $S_\mathrm{u} = T$ at equilibrium, which places the ratio of upper-ocean warming to GMST in b_u. Both S_u and T are anomalies from the 1995–2005 mean. Equation 9 shows that S_u is an average of past temperatures, weighted toward the recent past over a time window set by τ_u. The upper layer therefore smooths year-to-year fluctuations in GMST and follows longer trends with a delay comparable to τ_u.

The deep layer follows the same construction, with $\Delta\theta_\mathrm{d}(z,t) = S_\mathrm{d}(t)\,\varphi_\mathrm{d}(z)$ over 700–2000 m depth and $b_\mathrm{d} = \int_{h_\mathrm{u}}^{h_\mathrm{d}} \alpha_{0,\mathrm{sw}}\,\varphi_\mathrm{d}\,dz$, with $h_\mathrm{d} = 2000$ m. We drop the quadratic term because S_d lags well behind S_u ($\tau_\mathrm{d} \gg \tau_\mathrm{u}$). At the posterior median τ_u, S_d rises about 1.0

°C above its 1995–2005 level by 2100 in the 3.6 °C scenario, compared with about 3.0 °C for S_{u}, and the 21-year 0–2000 m record cannot constrain a second quadratic coefficient. Warming from GMST reaches the deep layer only after passing through the upper layer, so the thermal state S_{d} relaxes toward S_{u} as

$$\frac{dS_{\mathrm{d}}}{dt} = \frac{S_{\mathrm{u}} - S_{\mathrm{d}}}{\tau_{\mathrm{d}}}, \tag{10}$$

where we fix $\tau_{\mathrm{d}} = 150$ yr, the CMIP5 multi-model mean ratio of deep-ocean heat capacity to the heat-exchange coefficient in *Geoffroy et al.* (2013, Table 4), because the 21-year 0–2000 m data record is too short to constrain a centennial relaxation time. Adding this layer gives the total steric response for the upper 2000 m as

$$H_{\mathrm{ud}}(t) = H_{\mathrm{u}}(t) + b_{\mathrm{d}}\, S_{\mathrm{d}} + \epsilon_{\mathrm{u}}(t - t_0), \tag{11}$$

where the linear term $\epsilon_{\mathrm{u}}(t-t_0)$, with t_0 the first year of the GMST record, absorbs instrument drift during calibration and is omitted from the projections and the full-depth validation. Finally, to account for expansion below 2000 m depth, we define

$$H_{\mathrm{full}}(t) = H_{\mathrm{ud}}(t) + \dot{H}_{\mathrm{b}}(t - t_{\mathrm{ref}}), \tag{12}$$

where $\dot{H}_{\mathrm{b}} = 0.12 \pm 0.04\,\mathrm{mm\,yr^{-1}}$ (one standard deviation, 1991–2010) (*Desbruyères et al.*, 2016) is a constant rate for depths below 2000 m, referenced to year $t_{\mathrm{ref}} = 2000$. This layer is not observed at annual resolution, so its rate is adopted rather than calibrated.

We force the model with the Berkeley Earth GMST record (*Rohde and Hausfather*, 2020) and fit it jointly to thermosteric sea level estimates (*Levitus et al.*, 2012) at 0–700 m (1955–2025), compared with $H_{\mathrm{u}} + \epsilon_{\mathrm{u}}(t - t_0)$, and at 0–2000 m (2005–2025), compared with H_{ud} (Eq. 11). We use Bayesian inference (*Tarantola*, 2005) in which Markov chain Monte Carlo (MCMC) sampling draws parameter combinations $\boldsymbol{\phi} = (a_{\mathrm{u}}, b_{\mathrm{u}}, b_{\mathrm{d}}, \epsilon_{\mathrm{u}}, \tau_{\mathrm{u}}, \sigma_{\mathrm{extra}})$ with probability density proportional to the posterior (Fig. S2 of the Supplementary Material). Here, $r_i(\boldsymbol{\phi})$ is the residual at point i after subtracting each record's mean, and the model's mean over the same years, which removes each record's arbitrary reference level, and σ_i^2 is that point's total variance. Assuming the residuals are independent and normally distributed about the model, each with variance σ_i^2, the negative log-likelihood takes the form $\frac{1}{2}\sum_i [(r_i(\boldsymbol{\phi})/\sigma_i)^2 + \ln \sigma_i^2]$. The relaxation time of the upper layer, τ_{u}, has an informative prior, $\ln \tau_{\mathrm{u}} \sim \mathcal{N}(\ln 8, 0.5^2)$ with τ_{u} in years, so that $\mu_{\tau_{\mathrm{u}}} = \ln 8$ and $s_{\tau_{\mathrm{u}}} = 0.5$, where $\mathcal{N}$ denotes a normal distribution. The prior median is 8 yr and its 90% interval is [3.5, 18] yr, consistent with the ratio of upper-layer heat capacity to the heat-exchange coefficient, which spans 4 to 17 yr across the CMIP5 models in *Geoffroy et al.* (2013, Table 4), with a multi-model mean of 10 yr. Adding the negative log-prior for τ_{u} gives the objective function

$$\mathcal{J}(\boldsymbol{\phi}) = \sum_i \left[\left(\frac{r_i(\boldsymbol{\phi})}{\sigma_i} \right)^2 + \ln \sigma_i^2 \right] + \left(\frac{\ln \tau_{\mathrm{u}} - \mu_{\tau_{\mathrm{u}}}}{s_{\tau_{\mathrm{u}}}} \right)^2, \tag{13}$$

which is twice the negative log-posterior up to an additive constant. The $\ln \sigma_i^2$ term remains because σ_i depends on the sampled σ_{extra}. The point variance is defined by the quadrature

sum of the formal measurement error, a structural uncertainty equal to 5% of the bias-corrected anomaly, retained after a bias correction in the NOAA thermosteric observations (*Li et al.*, 2022), and σ_{extra}, a noise term, shared across both the 0–700 m and 0–2000 m records, that absorbs misfit beyond the reported observational error. The MCMC sampling uses the emcee package with 64 walkers, 6000 burn-in steps, and 30,000 production steps. The longest integrated autocorrelation time is 318 steps, about 1/94 of the chain length, giving at least 6000 effective samples per parameter. The parameters b_{u} and τ_{u} are partly degenerate in the likelihood, because a slower upper-layer response can be offset by a larger sensitivity to fit the same trajectory. This degeneracy is broken by τ_{u}'s informative prior. The other five parameters, a_{u}, b_{u}, b_{d}, ϵ_{u}, and $\ln \sigma_{\text{extra}}$, have flat priors, with a_{u}, b_{u}, and b_{d} constrained to be non-negative (Fig. S2). The posterior medians and 90% credible intervals are $a_{\text{u}} = 20.0$ [10.1, 31.2] mm $^{\circ}\text{C}^{-2}$, $b_{\text{u}} = 51.0$ [40.6, 75.4] mm $^{\circ}\text{C}^{-1}$, $b_{\text{d}} = 68.6$ [37.0, 104.3] mm $^{\circ}\text{C}^{-1}$, $\epsilon_{\text{u}} = 0.023$ [−0.112, 0.140] mm yr^{-1}, $\tau_{\text{u}} = 11.6$ [6.2, 25.3] yr, and $\sigma_{\text{extra}} = 2.0$ [1.7, 2.3] mm (Fig. S2). The calibrated model reproduces both records with RMS residuals of 2.3 mm for depths 0–700 m and 1.3 mm for 0–2000 m, against 42 and 30 mm of observed steric rise over the respective records (Fig. 3).

We validate the full-depth model in Eq. 12 against two global-mean steric sea level records, both withheld from calibration (Fig. 3c,f). *Frederikse et al.* (2020) reconstruct steric sea level for 1900–2018. Over 1957–2018 they draw, for each ensemble member, from three gridded products built on the same archive of in situ temperature and salinity profiles (*Levitus et al.*, 2012; *Ishii et al.*, 2017; *Cheng and Zhu*, 2016). The first of these is the data our model is calibrated on, so over this period the comparison measures consistency between separate analyses of largely shared data. Before 1957, in situ coverage is too sparse to resolve global steric change, and *Frederikse et al.* (2020) instead use the *Zanna et al.* (2019) reconstruction of ocean heat storage. That reconstruction is inferred from sea-surface temperature anomalies and heat-transport pathways in an ocean reanalysis. *Zanna et al.* (2019) also supplies their expansion below 2000 m over the full 1900–2018 period, whereas our model uses the constant rate of *Desbruyères et al.* (2016) for this layer. The pre-1957 segment is therefore the part of this comparison that is independent of the data used in calibration. Our model agrees with the reconstruction to an RMS difference of 2.9 mm over 1957–2018 and 2.7 mm before 1957. Our second validation comes from *Dangendorf et al.* (2024) who assimilate 516 tide-gauge records and satellite altimetry in an ensemble Kalman smoother that solves for GMSL over 1900–2021. Their global-mean steric estimate is what remains of GMSL after prescribed barystatic (from *Frederikse et al.*, 2020), glacial isostatic adjustment, and inverse-barometer contributions are removed. No temperature or salinity measurement enters this calculation, so the comparison tests our model against a steric estimate that shares none of the in situ ocean data used to calibrate it. Our model agrees with their reconstruction to an RMS difference of 3.1 mm over 1957–2021 and 6.0 mm before 1957 (Fig. 3c,f).

2.1.3 Glaciers

Projected 21st-century glacier mass loss scales approximately linearly with global warming (*Rounce et al.*, 2023; *Edwards et al.*, 2021). We therefore model the rate of the glacier contribution to sea level as linear in GMST,

$$\dot{H}_{\text{gl}} = b_{\text{gl}} T + c_{\text{gl}}, \tag{14}$$

where T is the calendar-year mean Berkeley Earth GMST anomaly relative to 1995–2005 (*Rohde and Hausfather*, 2020), b_{gl} is the sensitivity of the loss rate to warming, and c_{gl} is the loss rate at the 1995–2005 mean GMST. To treat uncertainty the same way as in the other components, we calibrate the model ourselves by Bayesian inference against the annual global glacier mass changes of the Glacier Mass Balance Intercomparison Exercise (GlaMBIE) for 2000–2023 (*The GlaMBIE Team et al.*, 2025). The GlaMBIE global total sums all 19 regions of the Randolph Glacier Inventory, including the peripheral glaciers of Greenland and Antarctica. Because the Greenland SMB emulator covers only the contiguous ice sheet (Section 2.2.1), the Greenland peripheral glaciers enter the projections only through this component. We fit the time-integrated form of Eq. 14, which gives the cumulative contribution,

$$H_{\mathrm{gl}}(t) = b_{\mathrm{gl}} \int_{t_0}^{t} T(t')\,\mathrm{d}t' + c_{\mathrm{gl}}(t - t_0) + H_{\mathrm{gl},0} + \varepsilon_{\mathrm{gl}}(t), \tag{15}$$

where $t_0 = 2000$ is the baseline year and $H_{\mathrm{gl},0}$ is the modeled contribution in that year. The observational error $\varepsilon_{\mathrm{gl}}$ is normally distributed with covariance Σ_{gl}. Summing annual mass changes into a cumulative record correlates its errors in time, and Σ_{gl} is the covariance of the cumulative record implied by independent annual errors. GlaMBIE reports its uncertainties as 95% intervals, which we convert to standard deviations. The residuals scatter more widely than these uncertainties allow, with a reduced chi-square of 4.1, so we add a model-inadequacy term $\sigma_{\mathrm{gl,extra}}$ in quadrature to each annual uncertainty and fit it with the other parameters.

We apply a weakly informative prior $b_{\mathrm{gl}} \sim \mathcal{N}(0.61,\, 2.0^2)$ mm yr^{-1} °C^{-1}, truncated to $b_{\mathrm{gl}} \geq 0$. The prior mean is drawn from *Rounce et al.* (2023), whose end-of-century rates of glacier sea-level rise increase from 0.70 to 2.23 mm yr^{-1} as warming above pre-industrial increases from 1.5 to 4 °C, a slope of 0.61 mm yr^{-1} °C^{-1}. The standard deviation is wide enough that the prior is only weakly informative, and widening it fiftyfold leaves the posterior unchanged. The non-negativity truncation recognizes that glaciers lose mass as the climate warms.

Left free, c_{gl} makes the fitted model gain glacier mass in almost every year before about 1970. In reality, glaciers entered the industrial era close to balance with the slowly cooling climate of the preceding centuries (*Roe et al.*, 2021). We therefore require the loss rate to be zero at a balance temperature $T_{\mathrm{gl,eq}}$, which gives $c_{\mathrm{gl}} = -b_{\mathrm{gl}} T_{\mathrm{gl,eq}}$. The model we fit is

$$H_{\mathrm{gl}}(t) = b_{\mathrm{gl}} \int_{t_0}^{t} [T(t') - T_{\mathrm{gl,eq}}]\,\mathrm{d}t' + H_{\mathrm{gl},0} + \varepsilon_{\mathrm{gl}}(t). \tag{16}$$

The prior on $T_{\mathrm{gl,eq}}$ is normal, centered on the 1850–1879 mean GMST of −0.86 °C relative to 1995–2005, with a standard deviation of 0.15 °C. Equation 16 neglects the adjustment of glacier geometry to climate, which sets each glacier's response time (*Roe et al.*, 2021), so it represents the limit of long response times.

The MCMC sampling uses the emcee package with 32 walkers, 2000 burn-in steps, and 4000 production steps. The posterior medians are $b_{\mathrm{gl}} = 0.76$ [0.60, 0.99] mm yr^{-1} °C^{-1}, $T_{\mathrm{gl,eq}} = -0.78$ [−1.03, −0.55] °C, $c_{\mathrm{gl}} = 0.59$ [0.52, 0.66] mm yr^{-1}, and $\sigma_{\mathrm{gl,extra}} = 0.18$ [0.13, 0.26] mm yr^{-1} (Fig. 4a). With c_{gl} free, the fit gives $b_{\mathrm{gl}} = 0.99$ [0.56, 1.44] mm yr^{-1} °C^{-1} and $c_{\mathrm{gl}} = 0.53$ [0.40, 0.66] mm yr^{-1}. The 2000–2023 record spans only about 0.8 °C of GMST, so

it constrains the present loss rate more tightly than the sensitivity. The balance constraint adds information from the pre-industrial climate, which lowers the median of b_{gl} by about a quarter and narrows its interval.

We validate the glacier model against the global glacier mass changes of *Zemp et al.* (2019) for 1962–1999, which are derived from glaciological and geodetic observations and are not used in the calibration. Over these 38 years the model gives a mean rate of 0.30 [0.17, 0.40] mm yr^{-1}, compared with an observed 0.32 [−0.02, 0.66] mm yr^{-1}, where the observed interval is the 90% range derived from the 95% uncertainties reported by *Zemp et al.* (2019) (Fig. 4a). Because the model with c_{gl} free predicts glacier mass gain before about 1970, this comparison also tests the balance constraint. We compare only the years before 2000. Over 2000–2016, when both records exist, the loss rate of *Zemp et al.* (2019) exceeds that of GlaMBIE by 0.18 mm yr^{-1}, so a comparison there would measure the disagreement between the two records rather than the skill of the model.

2.1.4 Antarctic Peninsula

The Antarctic Peninsula component uses the same level-space Bayesian framework as glaciers, fit at annual resolution to the IMBIE-3 Peninsula time series spanning 1979–2023 (45 annual points; *Otosaka et al.*, 2026). We choose the linear rate–temperature model $\dot{H}_{\mathrm{p}} = b_{\mathrm{p}} T + c_{\mathrm{p}}$, which captures the combined response of SMB and ice discharge to warming. IMBIE-3 attributes the Peninsula mass loss over 1979–2023 to ice discharge (*Otosaka et al.*, 2026), which increased after the Larsen A and B ice shelves collapsed in response to regional atmospheric warming (*Scambos et al.*, 2004). Here $H_{\mathrm{p},0}$ is the model level at the start of the record (1979), fitted with the other parameters, and the likelihood uses the covariance of this cumulative record implied by independent annual IMBIE-3 errors. As for glaciers, we also fit a model-inadequacy term $\sigma_{\mathrm{p,extra}}$ to account for annual scatter beyond the IMBIE-3 reported uncertainty, which lowers the reduced chi-square from 2.64 to 0.97. The MCMC sampling uses the emcee package with 32 walkers, 2000 burn-in steps, and 4000 production steps. The priors are $b_{\mathrm{p}} \sim \mathcal{N}(0,\, 2.0^2)$ mm yr^{-1} °C^{-1} and $c_{\mathrm{p}} \sim \mathcal{N}(0.3,\, 1^2)$ mm yr^{-1}. The posterior median sensitivity is $b_{\mathrm{p}} = 0.044$ [−0.022, 0.106] mm yr^{-1} °C^{-1}, and the rate at the 1995–2005 mean GMST is $c_{\mathrm{p}} = 0.056$ [0.039, 0.072] mm yr^{-1} (Fig. 4b). The 90% credible interval for b_{p} spans zero, so these data do not, on their own, resolve the sign of the temperature sensitivity.

2.1.5 East Antarctica

We fit the linear model $\dot{H}_{\mathrm{e}} = b_{\mathrm{e}} T + c_{\mathrm{e}}$ to the IMBIE-3 East Antarctic Ice Sheet (EAIS) record (1979–2023, 45 annual points; *Otosaka et al.*, 2026) in the same level-space framework as the Peninsula, with a weak prior $b_{\mathrm{e}} \sim \mathcal{N}(-0.17,\, 0.5^2)$ mm yr^{-1} °C^{-1} and no model-inadequacy term (reduced chi-square 0.94). The posterior median sensitivity is $b_{\mathrm{e}} = -0.043$ [−0.172, 0.088] mm yr^{-1} °C^{-1} (Fig. 4c). This interval spans zero, so the record does not constrain the sensitivity. The forecast instead adopts the snowfall-driven mass gain of $C_T = 60 \pm 20$ Gt yr^{-1} per °C of GMST (*Frieler et al.*, 2015; *Ligtenberg et al.*, 2013), which converts at 362.5 Gt per mm to −0.17 mm yr^{-1} °C^{-1} of sea level and also sets the prior mean (Section 2.2.1).

2.1.6 West Antarctica

Unlike for glaciers, the Antarctic Peninsula, and EAIS, we do not fit a rate–temperature regression to WAIS over the observational record, because ocean heat has driven most WAIS mass loss and ocean observations are too sparse to support a fit to ocean temperature. We instead fit a rate that changes linearly in time to the 1979–2023 IMBIE-3 record (*Otosaka et al.*, 2026), so the cumulative contribution is quadratic in time. As for the Antarctic Peninsula, the likelihood is that of the annual increments of the cumulative record, each with the reported IMBIE-3 uncertainty plus a fitted excess scatter. The acceleration a_{w} has a zero-mean prior, $a_{\mathrm{w}} \sim \mathcal{N}(0,\, 0.05^2)$ mm yr^{-2}. The fit yields an acceleration of 0.0122 [0.0100, 0.0144] mm yr^{-2} and a rate in year 2000 of 0.247 [0.220, 0.273] mm yr^{-1} (Fig. 4d). *Davison et al.* (2023) report snowfall anomalies in the Amundsen Sea Embayment over 2009–2013 and 2019–2020 that modulate this record. Removing them lowers the extrapolated contribution at 2100 from 86 to 79 mm, well within the S1 uncertainty described below, so we fit the record as reported. We set the WAIS rate to zero before 1979, when the IMBIE-3 record begins.

2.2 Forecasting methods

This section describes how the forecast joins the observed satellite-era trend to the component models, how uncertainty is propagated, and how each component is linked to projected GMST. The forecasts presented here use the IPCC AR6 GMST trajectories for different emission scenarios, though they can be applied to any GMST trajectory. We focus on the policy-relevant emission scenarios, labeled as in the Introduction by their AR6 central-estimate warming over 2081–2100: 1.8, 2.7, and 3.6 °C. We initialize the GMSL forecast using the GMSL level, rate, and acceleration observed in the satellite era and blend in the component-model sum in rate space, such that

$$\dot{H}_{\mathrm{forecast}} = w\dot{H}_{\mathrm{obs}} + \big[1 - w\big]\dot{H}_{\mathrm{comp}}, \tag{17}$$

where $\dot{H}_{\mathrm{obs}}$ is the rate of our quadratic fit to the NASA satellite altimetry record over 1993–2025, following *Nerem et al.* (2022) and *Hamlington et al.* (2024), and $\dot{H}_{\mathrm{comp}}$ is the rate of the summed component projections. The blending weight $w(t)$ declines smoothly from 1 to 0,

$$w = 1 - \Theta\left(\frac{t - t_{\mathrm{r}}}{\tau_{\mathrm{r}}}\right), \tag{18}$$

where $\Theta(x) = (1 + e^{-x})^{-1}$ is the standard logistic function, $t_{\mathrm{r}} = 2035$, and $\tau_{\mathrm{r}} = 5$ yr. At the forecast origin $w = 0.87$. At $t \ll t_{\mathrm{r}}$, $w \approx 1$ and the forecast follows the observed quadratic trajectory. At $t \gg t_{\mathrm{r}}$, $w \approx 0$ and the forecast follows the component models. The weight falls from 0.9 to 0.1 over $t_{\mathrm{r}} \pm 2.2\tau_{\mathrm{r}}$, a span of about 22 yr centered on 2035. We chose $t_{\mathrm{r}} = 2035$, 10 yr after the forecast origin, or about one third of the length of the 1993–2025 satellite altimetry record, a period when the extrapolation of observations should have strong forecasting skill. The tapering recognizes the expected reduction in forecasting skill of the simple data extrapolation over timescales longer than roughly half of the observational window. Moving t_{r} 5 yr later, to about half the length of the altimetry record and arguably

the latest viable choice, moves the 2050 and 2100 medians by 4 mm or less at 1.8–3.6 °C, so the results are insensitive to t_r between 2035 and 2040.

The GMSL forecast is obtained by integrating Eq. 17 forward from the observed level at the forecast origin ($t_f = 2025$, the last year of NASA altimetry data at the time of writing), taken from the smoothed NASA record with annual and semiannual cycles removed, so that

$$H_{\text{forecast}}(t) = H_{\text{obs}}(t_f) + \int_{t_f}^{t} \dot{H}_{\text{forecast}}(t')\, dt'. \tag{19}$$

Each member starts from the observed level shifted by its own quadratic fit's departure at t_f, so the starting level carries the 5.5 mm (1σ) uncertainty of the fit and stays consistent with that member's early rate. Blending rates rather than levels keeps the forecast continuous with the observed level at the forecast origin, so the offset between the component-sum level and the observed level does not enter the forecast. Each component has its own 2000-member ensemble of level-space projections $H_k(t)$ (for WAIS, a random subsample of a 10,000-member ensemble), with parameters drawn independently from that component's posterior or, for components using prescribed literature sensitivities rather than data calibration, from the corresponding parameter distributions. Member i of every GMST-driven component follows the same GMST path. This path is the AR6 median plus z_i times the growth of the AR6 half-width since 2024, taken toward the 5th or 95th percentile according to the sign of z_i, with z_i one standard-normal draw held for the whole century and used in every scenario. Through 2024, all members follow observed GMST. The total level is formed sample by sample, $H_{\text{comp},i}(t) = \sum_k H_{k,i}(t)$ for $i = 1, \ldots, 2000$, so parameter draws are independent across components while the within-scenario warming path is shared. The component rate $\dot{H}_{\text{comp}}$ used in Eq. 17 is obtained by finite-differencing this summed ensemble in time before blending with the observational quadratic rate. The blending is done member by member across the 2000-member ensemble. The quadratic coefficients are drawn jointly from their fitted covariance, independently of the components, so the correlations within the component sum, including the shared warming path, carry into the forecast.

2.2.1 Component-level GMST dependence

Four components need an extra step to link them to projected GMST. Greenland ice discharge depends on the temperature of the ocean around Greenland rather than on GMST. Greenland SMB is emulated from a regional climate model (Section 2.1.1.1). The EAIS sensitivity is not constrained by the observed record (Section 2.1.5), so we adopt a literature value. Terrestrial water storage is adopted from AR6.

To project the GrIS discharge model forward in time, we regress ocean temperature on Greenland regional surface temperature, a linear transfer function. We then use an Arctic amplification factor, the ratio of Greenland regional to global warming, drawn from a normal distribution, to calculate the Greenland regional surface temperature from projected GMST under different emission scenarios. The linear transfer function on annual-mean observations is

$$T_{\text{ocean}}(t) = \xi T^{\text{Gr}}_{\text{surface}}(t) + \chi + \varepsilon_{\text{go}}(t), \tag{20}$$

where $T^{\text{Gr}}_{\text{surface}}$ is the Greenland regional surface temperature anomaly and ε_{go} is a zero-mean residual. Both series are anomalies relative to the 1995–2005 baseline. An ordinary least-

squares fit over 52 common annual values gives $\xi = 0.148 \pm 0.022$ and $\chi = -0.058 \pm 0.045\,^{\circ}\mathrm{C}$ (1σ), with standard errors that account for the autocorrelation of the annual residuals and a correlation of 0.47 between the two parameters. The residual has standard deviation $\sigma_{\varepsilon_{\mathrm{go}}} = 0.20\,^{\circ}\mathrm{C}$ and lag-1 autocorrelation 0.48, and we carry it into the projections and into the hindcast before 1970 as first-order autoregressive noise.

For projections, the scenario GMST trajectories are converted to Greenland regional surface temperature using $\mathrm{AA} = 2.5 \pm 0.5$. The Arctic as a whole warmed nearly four times faster than the globe over 1979–2021 (*Rantanen et al.*, 2022), but the ratio of Greenland regional to global warming in the observations we use is lower and depends on the averaging window, spanning 2.1 to 2.7 for windows that start between 1970 and 1990 and end in 2024. Greenland regional surface temperatures are used in Eq. 20 to obtain projected T_{ocean}, which drives the lagged discharge model (Eqs. 1 and 2). Each of the 2000 Monte Carlo members draws $\mathrm{AA} \sim \mathcal{N}(2.5, 0.5^2)$ independently, (ξ, χ) jointly from their fitted covariance, and one autocorrelated path of $\varepsilon_{\mathrm{go}}$, together with one draw of (γ, r_0, δ) from the discharge posterior, so that Arctic amplification, transfer-function, and discharge parameter uncertainties all propagate into the projections. A single draw of $(\mathrm{AA}, \xi, \chi, \gamma, r_0, \delta)$ is applied to all scenarios, so that differences between scenarios isolate the effect of the emissions pathway from parameter uncertainty.

For EAIS SMB, we use the linear model of Section 2.1.5 with the literature-derived accumulation sensitivity $C_T = 60 \pm 20$ Gt yr^{-1} °C^{-1} of GMST (*Frieler et al.*, 2015; *Ligtenberg et al.*, 2013) (about 5% per °C applied to EAIS accumulation, taking Antarctic warming equal to GMST), equivalent to $b_{\mathrm{e}} = -0.17$ mm yr^{-1} °C^{-1}. This sensitivity reflects the Clausius–Clapeyron scaling of snowfall with warming ($C_T > 0$ corresponds to EAIS mass gain through accumulation). Data from IMBIE-3 show EAIS in approximate balance (Fig. 4c), so we set $c_{\mathrm{e}} = 0$. The EAIS contribution is small (-14 to -25 mm at 2100 across the 1.8–3.6 °C scenarios) relative to the total, meaning the GMSL projection results of this study are largely insensitive to these parameter choices.

Terrestrial water storage (TWS) projections are adopted directly from IPCC AR6 rather than modeled here. Groundwater depletion, reservoir impoundment, and changes in land hydrology are outside the scope of this framework, and the data are too sparse for a data-driven model. At each IPCC time step we fit a normal distribution to the reported 5th, 17th, 50th, 83rd, and 95th percentiles. Each of the 2000 members takes a single quantile of these distributions for the whole century, and its path is linearly interpolated to annual resolution. For the hindcast and budget-closure diagnostics we use the *Frederikse et al.* (2020) TWS estimate only from 2003 onward, with Monte Carlo paths whose errors accumulate from 2003 so that each path is continuous in time. Their TWS is derived from GRACE and GRACE-FO observations over 2003–2018, partitioned from the total land-mass signal. Before 2003 it is a composite of a reconstruction of climate-driven water storage and process models for reservoir impoundment and groundwater depletion rather than a measurement. We therefore set TWS to zero before 2003, rebased so the cumulative contribution is continuous across the transition, and we note that the pre-2003 component sum consequently omits a real contribution to sea level. TWS varies little among scenarios and its contribution is small relative to the total, so it does not materially affect the uncertainty budget.

2.2.2 West Antarctic Ice Sheet

West Antarctica dominates the uncertainty in sea-level projections because its major outlet glaciers, particularly Thwaites and Pine Island, lie close to a threshold at which small changes in ice geometry could trigger self-sustaining retreat (*Joughin et al.*, 2014; *Pattyn and Morlighem*, 2020; *Fricker et al.*, 2025). Whether that threshold has been, or will soon be, crossed depends on glacier-specific details that remain only partly observed. We do not attempt to constrain a GMST sensitivity for WAIS. Instead, we represent WAIS as a mixture of two scenarios. In Scenario 1 (S1), mass loss continues along the observed trend. In Scenario 2 (S2), loss accelerates well beyond it.

Given growing evidence that WAIS mass loss is primarily sensitive to internal ice dynamics and to deep-ocean temperatures whose 21st-century warming is largely committed regardless of emissions, we make our WAIS projections independent of the warming trajectory (*Naughten et al.*, 2023; *Coulon et al.*, 2024; *Rosier et al.*, 2025). This independence is limited to the 21st century, because warming over this century largely sets the Antarctic contribution in the centuries that follow (*Lowry et al.*, 2021; *Seroussi et al.*, 2024). Some studies find emissions dependence even before 2100, with major WAIS retreat beginning this century in some simulations under 3 °C of warming but not under 2 °C (*DeConto et al.*, 2021). An emulation of the ISMIP6 Antarctic ensemble extended to 2300, calibrated to the observed 2002–2021 Antarctic mass-loss rate, finds that higher emissions raise the 2100 Antarctic contribution with probability 0.89, but its posterior median for the whole ice sheet rises by only 5 mm from the 1.4 to the 2.7 °C scenario (*Lin et al.*, 2026). That shift is small compared with our S1 90% credible interval of 86 mm [−10, 182], and part of it arises from the SMB response that we carry separately for EAIS.

In S1, mass loss continues along the observed trajectory (Fig. 4d). Extrapolating the trend gives a median GMSL contribution of 86 mm by 2100, 1 mm above the ISMIP6 emulator median of 85 mm (*Edwards et al.*, 2021). A 45-year observational record cannot itself constrain how much additional error should accumulate from extrapolating a further 76 years beyond it, so a credible interval built from the fitted parameters alone would understate the uncertainty in this extrapolation. We instead treat the growth in uncertainty that the ISMIP6 emulator reports beyond the last observation as an external estimate of the extrapolation error. We add it to our parameter uncertainty as a prediction error (*Tarantola*, 2005), which widens our 90% credible interval at 2100 to [−10, 182] mm, closely matching the ISMIP6 emulator range of [−11, 182] mm.

In S2, mass loss accelerates well beyond currently observed rates. We define the S2 distribution at 2100 by its 5th percentile of 86 mm, equal to the S1 median, and its 95th percentile of 1.3 m, approximately the IPCC AR6 low-confidence Antarctic Ice Sheet 95th percentile at 2100 under the 4.4 °C scenario (*Fox-Kemper et al.*, 2021). The distribution is right-skewed. Its logarithm follows a skew-normal distribution with shape parameter $\lambda = 3$, reflecting the expectation that instability skews outcomes toward greater loss (*Robel et al.*, 2019). Varying λ from 0 to 6 moves the 2100 median of the mixture with $p(\mathrm{S1}) = 10\%$ from 0.30 to 0.20 m and its 95th percentile by 0.01 m. The trajectory of WAIS mass loss through time is set by a power law in time that starts at the observed rate and accelerates toward the 2100 value, with median exponent $\beta = 2$, for which the S2 95th percentile stays within the AR6 medium-confidence range for the Antarctic Ice Sheet until about 2050 and exceeds

it afterward.

Emerging research on physical processes not yet represented in ice-sheet models supports the possibility of loss well above the observed trajectory. Examples include seawater intrusion beneath grounded ice (*Rignot et al.*, 2024; *Bradley and Hewitt*, 2024) and the evolution of subglacial water pressure (*Zhao et al.*, 2025). The ISMIP6 ensemble itself implies far less loss. *Lin et al.* (2026) identify a set of ice-sheet model choices that remain consistent with the observed mass-loss rate and yield 157 mm [98, 254] for the whole ice sheet at 2100 under the 4.4 °C scenario. The 95th percentile of that range is roughly a fifth of our S2 95th percentile for WAIS alone. That ensemble largely omits marine ice cliff instability, subglacial hydrology, and ice damage, so the difference reflects processes absent from current ice-sheet models. The S2 95th percentile adopted here should therefore not be read as an upper bound on plausible loss, and Fig. 6 shows how the WAIS contribution changes with it.

The final probability density is a simple mixture of S1 and S2, weighted by the probability of S1, $p(\mathrm{S1})$ (Fig. 5). Its median, 95th percentile, and other quantiles are determined primarily by $p(\mathrm{S1})$ and the S2 95th percentile (Fig. 6). We hold the S2 95th percentile fixed and use $p(\mathrm{S1})$ as the control variable. We present two storylines at the limits of the IPCC terms "very likely" (90%) and "very unlikely" (10%): a slow-WAIS storyline with $p(\mathrm{S1}) = 90\%$, in which continued mass loss along the observed trajectory is very likely, and a fast-WAIS storyline with $p(\mathrm{S1}) = 10\%$, in which it is very unlikely.

3 Results

We first test the framework against observations and then present projections. Hindcasting skill matters for judging a model's projections, because it tests forecasting skill against data and indicates the horizon over which the projections can be trusted. We test the summed components against satellite altimetry (budget closure) and against GMSL reconstructions back to 1900 (hindcasts), then present projections for each component and for total GMSL.

3.1 Validation

We validate the model in two ways. First, we test whether the sum of the separately calibrated components accounts for the observed rate of GMSL rise (budget closure). Second, we run the model over the past century with observed GMST (a hindcast) and compare it with independent GMSL records.

3.1.1 Total GMSL budget closure

Satellite altimetry over 1993–2025 provides observations of GMSL (*Beckley et al.*, 2017; *Hamlington et al.*, 2020) that can be compared with the sum of the components (Fig. 7). The thermosteric, glacier, and Peninsula contributions come from the calibrated models driven by observed GMST, and Greenland discharge from its model driven by observed ocean temperature. WAIS and EAIS enter as the IMBIE-3 records (*Otosaka et al.*, 2026) through 2023, continued by the WAIS fit and the EAIS SMB model. Total GMSL is withheld

from calibration to serve as a validation data set. For Greenland SMB we use the RACMO-derived SMB of *Mouginot et al.* (2019) through 2018 and the emulator thereafter. Terrestrial water storage is adopted from *Frederikse et al.* (2020) from 2003 onward, the period over which their estimate derives from GRACE observations, set to zero before, and held at its 2018 value after their record ends. The uncharted-glacier term, described in the hindcast subsection below, is zero from 2000 onward.

The component sum closes 94–103% of the observed rate of GMSL rise over three epochs (1993–2025, 2000–2025, 2010–2025), indicating that the individually calibrated components together account for observed GMSL during the satellite altimetry record (Fig. 7). Because each component is calibrated independently, we treat their rate uncertainties as independent and combine them in quadrature. Each component's individual uncertainty comes directly from its Monte Carlo ensemble. For a given epoch, we fit a linear trend to every ensemble member and take the standard deviation of the resulting trends. Repeating this for each epoch lets the combined uncertainty differ from one epoch to the next, rather than using one fixed value throughout the satellite record. We add the uncertainty of the altimetry trend over the same epoch in quadrature, combining the published altimetry error covariance, the serial correlation of the residuals, and the uncertainty of the glacial isostatic adjustment correction. Errors on the cumulative records that enter the closure directly, IMBIE-3 for WAIS and EAIS, the observed Greenland SMB, and terrestrial water storage, are drawn as correlated annual increments. By this measure, the residual between the observed and summed rates is 0.36σ, 0.60σ, and 1.26σ of the combined uncertainty for 1993–2025, 2000–2025, and 2010–2025. The largest shortfall is over 2010–2025. It arises after 2018, when the closure uses the SMB emulator in place of the observed Greenland SMB and holds terrestrial water storage at its 2018 value.

3.1.2 Hindcasting the level, rate, and transient sensitivity

We validate the combined GMST-sensitive model by two hindcasting approaches. In both, the GMST-dependent components are forced with observed GMST back to 1900 (*Rohde and Hausfather*, 2020), with an emphasis on post-1950 when increasingly reliable GMSL data are available. WAIS follows its fit to the IMBIE-3 record, and terrestrial water storage follows *Frederikse et al.* (2020) from 2003. The glacier hindcast also includes glaciers that are missing from the inventories on which the GlaMBIE record is built, because they are too small to map or because they melted away after 1901. *Parkes and Marzeion* (2018) estimate that these uncharted glaciers contributed 0.17 to 0.53 mm yr^{-1} over 1901–1990, most of it early in the century, and 0.08 to 0.21 mm yr^{-1} over 1993–2010. We add this contribution as a prescribed series, sampled between their lower and upper bounds. Within each bound the rate declines linearly over 1901–1990, with a mean equal to their 1901–1990 value, and then holds their 1993–2010 rate. It does not depend on GMST and is set to zero from 2000, where GlaMBIE calibrates the glacier model.

The first comparison is between the GMSL level and rate from SPECTRUM and those of the *Frederikse et al.* (2020) and *Dangendorf et al.* (2024) reconstructions, the IPCC AR6 compilation (*Fox-Kemper et al.*, 2021), which includes *Frederikse et al.* (2020) and an earlier version of the *Dangendorf et al.* (2024) reconstruction, and satellite altimetry (*Beckley et al.*, 2017). For the second comparison, we calculate the transient sensitivity of the SPECTRUM

output, the change in the rate of GMSL rise per degree of warming, $\alpha = d\dot{H}/dT$, where $\dot{H}$ is the rate of GMSL rise. For the level and rate comparison, we use the same rate-space blending scheme as the forecast (Section 2.2), reflected about the forecast origin so that the satellite-era quadratic dominates at late times and the component models at early times. The midpoint of the blend falls at 2015, the mirror of 2035 about $t_{\mathrm{f}} = 2025$, with $\tau_{\mathrm{r}} = 5$ yr unchanged. The observational term thus carries the same weight at 2025 as in the forecast and falls to 0.05 by 2000, so the hindcast over the reconstruction era tests the component models rather than restating the altimetry extrapolation.

Extending the hindcast back to 1900 shows distinct regimes relative to the various GMSL time series (Fig. 8a). Over the satellite era the hindcast tracks the observations closely, with a 1993–2025 rate of 3.24 mm yr^{-1} compared with 3.35, 3.17, and 3.23 mm yr^{-1} for the *Frederikse et al.* (2020) and *Dangendorf et al.* (2024) reconstructions and the IPCC AR6 observational compilation (*Fox-Kemper et al.*, 2021), and with the satellite altimetry record (*Beckley et al.*, 2017). Over 1950–2000 the hindcast rate is 1.11 mm yr^{-1} compared with 1.02, 1.01, and 1.28 mm yr^{-1} for the same three records, and the hindcast level at 1950 lies within 10 mm of all three. Of the hindcast rate, 0.21 mm yr^{-1} comes from the prescribed uncharted-glacier term. Because terrestrial water storage is set to zero before 2003, this comparison tests the land-ice and thermosteric components alone. Before 1950, where only the *Frederikse et al.* (2020) and *Dangendorf et al.* (2024) reconstructions extend, the divergence grows: the hindcast rate is 0.77 mm yr^{-1} compared with 1.39 and 1.64 mm yr^{-1} over 1900–1950, and the hindcast level at 1900 is 39–45 mm above the reconstructions, relative to the 2000 baseline. Before 1950, the median line from SPECTRUM falls outside the 90% interval of *Dangendorf et al.* (2024) only in 1906 and 1908. This pre-1950 divergence follows from a design choice. We include only model representations that can be fit to observations and that describe the ongoing response to warming. Representing GMSL in earlier, colder periods would require more complete land-ice process models, which are outside our scope. We therefore treat the pre-1950 hindcast as qualitatively informative but do not draw quantitative conclusions from it. We restrict our validation claims to the 75-year, post-1950 period.

Figure 8b compares the rate of GMSL rise of our component model with observed rates as a function of GMST, with both rates estimated by the same 30-yr local regression. The color of the model line shows its transient sensitivity along the 1957–2025 trajectory. The slope of the observed rate against GMST also agrees with the sensitivity of 3.4 mm yr^{-1} °C^{-1} found by *Rahmstorf* (2007) (Fig. 8b) nearly 20 years ago.

Together, the level, rate, and transient-sensitivity comparisons indicate that our framework reproduces the observed behavior of GMSL rise over a 75-year window (1950–2025) spanning 1.1–1.2 °C of GMST change (11-yr means), without having been calibrated on observations of GMSL itself. The hindcast reproduces the satellite-era rate (3.24 mm yr^{-1} over 1993–2025, compared with 3.17–3.35 mm yr^{-1} in the observational records). Its 1950–2000 rate of 1.11 mm yr^{-1} lies between those of the two tide-gauge reconstructions (1.01 and 1.02 mm yr^{-1}) and the 1.28 mm yr^{-1} of the IPCC AR6 compilation. Its transient sensitivity follows the observed increase in the rate of rise with warming.

This validation window is comparable in duration to the remaining forecast horizon, and in warming to the 2.7 °C scenario, which adds 1.4 °C between 2025 and 2100. The 1.8 °C scenario adds 0.4 °C, less than the warming across the validation window, although it takes GMST above any observed value, while the 3.6 °C scenario adds 2.6 °C, so its forecast

extends furthest beyond what the hindcast tests. This validated envelope does not, however, extend to processes for which comparable historical constraints do not exist. WAIS, the largest source of forecast uncertainty, is therefore represented in this work by scenario-based storylines rather than a data-driven model (Section 2.2.2).

3.2 Component projections

We drive the calibrated model components with the IPCC AR6 GMST trajectories for the 1.8, 2.7, and 3.6 °C scenarios (Fig. 9a). Current policies imply a median warming of 2.7 °C by 2100, with emissions close to or slightly below those of the 2.7 °C scenario, although warming of 4 °C or more cannot be fully ruled out (*Hausfather*, 2025). The 1.8 and 3.6 °C scenarios bracket current policies. Projected levels are for 2100 relative to the study baseline year 2000, given as the posterior median with the 90% credible interval in brackets. We compare each component with the IPCC AR6 medium-confidence projections, which AR6 builds from processes that can be modeled with confidence. AR6 reports projections relative to the 1995–2014 mean (*Fox-Kemper et al.*, 2021). For these comparisons we subtract each ensemble member's own 1995–2014 mean, and we call the result the AR6 baseline.

The thermosteric contribution is 148 [109, 207], 249 [186, 342], and 378 [274, 541] mm at 1.8, 2.7, and 3.6 °C, respectively (Fig. 9b). On the AR6 baseline, our medians are 142, 243, and 372 mm, which are 1.00, 1.23, and 1.49 times the AR6 medians for thermal expansion of 142, 197, and 249 mm. The quadratic term in the upper-layer expansion of the two-layer model (*Geoffroy et al.*, 2013) drives that divergence, because it makes each added degree of warming expand the upper ocean more than the last. In the fit, this term mainly captures departures from a linear response in the calibration record, such as warming that reaches deeper over time, with a smaller part from the temperature dependence of the thermal expansion coefficient of seawater. The deep layer adds a similar amount in all three scenarios, consistent with observed deep-ocean warming (*Cheng et al.*, 2017), so it raises all three projections without driving the divergence.

The Greenland Ice Sheet contributes 71 [35, 107], 93 [51, 132], and 117 [71, 163] mm at 1.8, 2.7, and 3.6 °C (Fig. 9c). On the AR6 baseline, our medians are 68, 90, and 114 mm, which are 1.24, 1.08, and 1.00 times the AR6 medians of 55, 83, and 114 mm. The calibrated ice-sheet model ensemble of *Aschwanden and Brinkerhoff* (2022) gives an independent comparison. Relative to 2008, when their simulations begin, it projects 80 [40, 140] mm under Representative Concentration Pathway (RCP) 2.6 and 110 [50, 180] mm under RCP4.5. Relative to the same year, we project 67 [30, 103] and 88 [47, 128] mm at 1.8 and 2.7 °C. Our Greenland discharge model is calibrated with the observed discharge record (*Mouginot et al.*, 2019), with two other mass-balance records withheld for validation (*Mankoff et al.*, 2021; *Otosaka et al.*, 2026). Our SMB emulator is trained on the MAR regional climate model alone. Among the regional climate models compared by *Glaude et al.* (2024), MAR is one of two that give about twice the SMB loss of RACMO under the same forcing.

We project glacier contributions of 121 [104, 143], 147 [125, 179], and 170 [141, 213] mm at 1.8, 2.7, and 3.6 °C (Fig. 9d). On the AR6 baseline, our medians are 118, 144, and 167 mm, which are 1.32, 1.17, and 1.06 times the AR6 medians of 89, 123, and 157 mm, so our glacier projection converges toward the AR6 median as warming increases. The GlacierMIP2 ensemble of 11 glacier models (*Marzeion et al.*, 2020) reports glacier mass loss over 2015–

2100. Its medians and 90% intervals are 79 [23, 135] mm under RCP2.6, 119 [53, 185] mm under RCP4.5, and 159 [73, 245] mm under RCP8.5. Our glacier contribution in 2015 is 10.0 mm above the 2000 baseline, and over 2015–2100 we project medians of 111, 137, and 160 mm at 1.8, 2.7, and 3.6 °C. The first two lie within the GlacierMIP2 intervals for RCP2.6 and RCP4.5, respectively, and the third is within 1 mm of the RCP8.5 median. We model the rate of glacier mass loss as linear in GMST, following glacier model ensembles (*Marzeion et al.*, 2020; *Edwards et al.*, 2021), and calibrate it with the GlaMBIE record (*The GlaMBIE Team et al.*, 2025). The prior on the glacier sensitivity b_{gl} is weakly informative, and its posterior is set by the GlaMBIE record together with the pre-industrial balance constraint.

The combined Antarctic Peninsula and EAIS contribution stays near zero, with medians of −3.8 [−13.7, 4.1], −8.0 [−22.1, 3.1], and −11.8 [−29.8, 2.4] mm at 1.8, 2.7, and 3.6 °C (Fig. 9e). Negative values denote sea-level fall. All three 90% intervals span zero, so the sign of the net contribution is not resolved. The EAIS contribution becomes more negative across the three scenarios, from −13.1 to −24.1 mm, as warming increases snowfall. An Antarctic Peninsula contribution that rises from +9.1 to +11.9 mm partly offsets this. The EAIS projection adopts an SMB sensitivity derived from published accumulation scalings, $C_T = 60 \pm 20$ Gt yr^{-1} per °C of GMST (*Frieler et al.*, 2015; *Ligtenberg et al.*, 2013). The Peninsula sensitivity is calibrated with the IMBIE-3 Peninsula record (*Otosaka et al.*, 2026). It captures the combined response of SMB and ice discharge to warming, including glacier acceleration after ice-shelf collapse (*Scambos et al.*, 2004). West Antarctica is treated separately (Section 2.2.2).

3.3 Full model projections

We report the forecast first under the fast-WAIS storyline ($p(\mathrm{S1}) = 10\%$; Section 2.2.2). For the 1.8 °C scenario, this forecast shows a median GMSL rise of 0.60 [0.41, 1.59] m relative to 2000 (Fig. 10a). After referencing our projection to the AR6 baseline, this median is 35% above the IPCC AR6 medium-confidence median of 0.44 m. Under 2.7 °C warming, approximately the current trajectory (*Hausfather*, 2025), the median is 0.75 [0.54, 1.74] m, 33% above the AR6 median of 0.56 m. For the 3.6 °C scenario, the median reaches 0.94 [0.67, 1.92] m, 37% above the AR6 median of 0.68 m. For reference, the 4.4 °C scenario yields a median of 1.13 [0.78, 2.12] m.

Relative to pre-industrial GMSL (adding ∼0.15 m, the 0.14 m of observed rise from 1900 to 2000 (*Frederikse et al.*, 2020; *Dangendorf et al.*, 2024) plus ∼0.01 m from the 1850–1900 mean to 1900 (*Kopp et al.*, 2016; *Fox-Kemper et al.*, 2021)), these forecasts correspond to medians of 0.75–1.09 m of GMSL rise across the 1.8–3.6 °C policy-relevant scenarios. The probability of exceeding 1 m relative to pre-industrial by 2100 is 23% under 1.8 °C of warming, 34% under 2.7 °C, and 69% under 3.6 °C. The probability of exceeding 2 m relative to pre-industrial by 2100 ranges from 3 to 6% across the 1.8–3.6 °C scenarios and is lowest at 1.8 °C. Under the slow-WAIS storyline, the probability of exceeding 1 m relative to pre-industrial by 2100 falls to 3% at 1.8 °C, 4% at 2.7 °C, and 31% at 3.6 °C, and the probability of exceeding 2 m stays below 1% across the 1.8–3.6 °C scenarios (Fig. 10b). For comparison, our slow-WAIS forecast for 2.7 °C of warming is 23 mm below the extrapolation of the observed GMSL trend. This agreement is expected if warming continues along its current trend and WAIS, which contributed 9% of observed GMSL rise from 1993 to 2023,

continues to lose mass at its current rate.

Fig. 11a shows the mean 21st-century contribution of each component. Under the fast-WAIS storyline, WAIS governs the upper tail of the forecast distribution in every warming scenario (Fig. 10).

The distribution of the uncertainties shows a clear separation between the predictable components and those that are currently unpredictable (Fig. 11b). These results show that WAIS accounts for more than 90% of the variance, and 73 to 85% of the 90% credible interval, in 21st-century sea-level rise. The thermosteric, glacier, and Greenland contributions, all of which have calibrated sensitivities to GMST, contribute 1 to 6% to the total variance and 15 to 27% to the 90% credible interval, with their share of that interval growing as warming increases. As a result, the 90% uncertainty range of 1.20 m for 2.7 °C is 3.5 times the spread across warming scenario medians (0.34 m between 1.8 and 3.6 °C). This picture shifts under the slow-WAIS storyline. Those three components then set 43 to 61% of the 90% credible interval, up from 15 to 27% (right column of Fig. 11b). Because the WAIS forecast does not depend on GMST before 2100, sea-level risk separates into two dimensions, one reducible by emissions cuts and one set by WAIS, each addressed by a distinct class of societal response.

4 Discussion

4.1 Data-driven forecasts as part of a hierarchy

The SPECTRUM framework occupies an intermediate level of a hierarchy of sea-level models ordered by complexity, so its results can be compared with both simpler and more complex models. The simplest models extrapolate the observed trends. Extending our quadratic fit to the satellite altimetry record of GMSL to 2100, following *Nerem et al.* (2022) and *Hamlington et al.* (2024), who stop at 2050, gives 0.63 [0.38, 0.86] m of rise relative to 2000. At the local scale, trends and accelerations at 222 tide gauges over 1970–2023, extrapolated to 2050, are consistent with the AR6 medium-confidence projections within the likely range at 96% of locations (*Wang et al.*, 2025). Next in complexity are the semi-empirical models, which relate the rate of global mean sea-level rise to GMST. Applying the semi-empirical relation of *Rahmstorf* (2007) to the 2.7 °C scenario gives 0.73 [0.67, 0.82] m relative to 2000, where the interval reflects only the spread of warming paths. Our slow-WAIS forecast ($p(\mathrm{S1}) = 90\%$) gives 0.60 [0.47, 0.83] m at 2.7 °C (Fig. 10a). Its median is 23 mm less than the extrapolation of GMSL and 0.13 m less than the semi-empirical model. The agreement to within 0.13 m between the slow-WAIS forecast at 2.7 °C, which approximates the current warming trend, and the simpler approaches is a consistency check on the forecast.

At the complex end of the hierarchy are the process-model assessments of IPCC AR6 (*Fox-Kemper et al.*, 2021), which report two sets of projections. The medium-confidence set is built from processes that can be modeled with confidence, while the low-confidence set admits ice-sheet processes such as marine ice cliff instability along with expert judgment. In SPECTRUM, $p(\mathrm{S1})$ moves the forecast from the medium-confidence set toward the low-confidence set. To compare with AR6, we shift our forecast to the AR6 1995–2014 baseline, which lowers each value by 12 mm. On that baseline, our slow-WAIS forecast gives 0.45 [0.33, 0.64] m at 1.8 °C, 0.59 [0.46, 0.82] m at 2.7 °C, and 0.77 [0.60, 1.06] m at 3.6 °C. The

AR6 medium-confidence medians are 0.44, 0.56, and 0.68 m. Our slow-WAIS medians are 3 to 13% greater than those values, compared with 33 to 37% for the fast-WAIS forecast. Our slow-WAIS 95th percentiles are less than the AR6 medium-confidence 95th percentiles at all three warming levels (0.64, 0.82, and 1.06 m, compared with 0.78, 0.95, and 1.11 m). At 1.8 and 2.7 °C, most of the difference between our fast-WAIS forecast and the AR6 medium-confidence projections is a difference in the weight given to rapid WAIS mass loss. At 3.6 °C, the thermosteric, glacier, and Greenland contributions together account for about half of the difference, mostly through the thermosteric term. Our fast-WAIS forecast is greater than the AR6 low-confidence projections, which AR6 provides for the 1.8, 2.7, and 4.4 °C scenarios, except in the upper tail at 4.4 °C. At 2.7 °C, our median of 0.74 m and 95th percentile of 1.73 m are greater than the low-confidence values of 0.57 and 1.18 m. At 4.4 °C, our median of 1.11 m is greater than the low-confidence median of 0.88 m, while our 95th percentile of 2.11 m is less than their 2.28 m.

4.2 Completeness of the uncertainty budget

Aschwanden et al. (2021) separate the uncertainty in an ice-sheet forecast into five parts: the emissions scenario, the model structure, the initial state, the model parameters, and the irreducibly random component of the forcing. Their criticism of the ISMIP6 ensemble is that it integrated over model structure but not over parameters, in a setting where the parametric spread is the larger of the two. SPECTRUM does the converse. It integrates over parameters for every calibrated component, and over structure for the one component where structure dominates, through the WAIS scenario mixture. Within-scenario forcing uncertainty, the spread of GMST trajectories under a given emissions pathway, is sampled explicitly as a warming path shared by all components in each ensemble member. Initial-state and random-forcing uncertainty enter through the calibrated error models rather than being separated. What the intervals do not carry is structural uncertainty in the calibrated components themselves, in the sense of alternatives to the functional form each component takes. For the fast-WAIS forecast this has little consequence, because WAIS accounts for more than 90% of the variance. Under the slow-WAIS storyline, the calibrated components account for 43 to 61% of the summed component confidence interval widths, and their contribution to the total confidence interval is conditional on the model forms adopted here. Two of those choices have been tested directly. Replacing the adopted Arctic amplification factor with either end of the observed Greenland-to-global warming ratio (2.1 and 2.7), and shifting the 200–500 m ocean-forcing layer 100 m shallower or deeper, each move the 2100 median by 3 mm or less and the width of the 90% credible interval by 2 mm or less. *Lin et al.* (2026) integrate over the ice-model choices that ISMIP6 documented but did not sample. They emulate the ISMIP6-2300 ensemble as a function of 18 ice-dynamical features together with the choice of climate model, then place a uniform prior over those choices. In their raw simulations these features account for 64% of the 2015–2100 variance, with the sliding law contributing 21% and the ice-shelf melt parameterization 20%, compared with 9% for the choice of atmosphere-ocean general circulation model and 6% for GMST. The spread arising from choices made within the models is therefore several times the spread arising from the forcing.

We do not calculate the integral over model structure itself, in which a projection is

averaged over a distribution across model structures (*Aschwanden et al.*, 2021). That distribution is defined by the requirement that the true data-generating process lie within its support. Therefore the distribution cannot be recovered by assigning weights to the models that have been built and requiring those weights to sum to one. Doing so treats each model as an independent sample of the ways the system might be represented, when in fact those models describe the same system, rely on many of the same assumptions, and are tested against many of the same observations. What the SPECTRUM framework does implement is the conditioning step of *Aschwanden et al.* (2021), in which a prior over projections is filtered by observations. Observed GMSL is withheld from every component calibration, so the hindcast tests the assembled framework against a record none of its components saw. Over 1950–2000 the summed components rise at 1.11 mm yr^{-1}, compared with 1.02 and 1.01 mm yr^{-1} in the *Frederikse et al.* (2020) and *Dangendorf et al.* (2024) GMSL data sets and 1.28 mm yr^{-1} in the IPCC AR6 compilation. *Aschwanden et al.* (2021) point out that it is not yet settled how long a model should be required to match observations before it is trusted. The test used here covers the 75 years since 1950, which is longer than the mass-change records such comparisons usually rely on, although the reconstructions it is compared against also become less certain further back in time. *Lin et al.* (2026) take the alternative approach for Antarctica, assigning probabilities to the ISMIP6 modeling options and updating them against the observed mass-loss rate. Their result illustrates the limit of that approach as well as its value. The observations move the sliding-law probabilities, but for most other choices the posterior stays close to the prior, because a two-decade mass-balance record weakly constrains processes that have little effect on the recent trend and a large effect on the century-scale response. They describe the residual as strong ambiguity and address it with probability boxes over alternative priors, the same difficulty that leads us to represent WAIS with a scenario mixture rather than a single distribution.

4.3 Two dimensions of sea-level risk

The potential impact and uncertainty around WAIS mass loss creates two dimensions of risk. The first risk dimension is sensitive to emissions reduction. The temperature-dependent components are sensitive to changes in GMST. Here, we calibrate the sensitivities using decades of observations and validate the fit against decades of observations withheld from the calibration. The collective contribution of the calibrated components shifts systematically across warming trajectories and is reducible by emissions cuts (Fig. 11a). Under the slow-WAIS storyline, the 90% credible interval of the forecast spans 0.31 m at 1.8 °C, 0.36 m at 2.7 °C, and 0.47 m at 3.6 °C, 2.7 to 3.8 times narrower than the fast-WAIS forecast, which admits rapid mass loss from WAIS driven by internal feedbacks (Fig. 10). Emissions policy becomes the primary lever for managing risk in that world. Moving from 1.8 °C to 3.6 °C adds 0.32 m of rise, exposing 53 million more people to coastal flood risks and adding US$43B/yr in coastal adaptation costs.

The second risk dimension is relatively insensitive to emissions reductions over the 21st century. WAIS dynamics accounts for more than 90% of the total forecast variance and is largely decoupled from the emission pathway. The tail of the forecast distribution—the outcomes that drive the most severe human and economic consequences—is controlled almost entirely by whether WAIS retreat follows modern loss trends or transitions to a more rapid,

irreversible retreat (*Turney et al.*, 2020; *Dutton et al.*, 2015; *Graham et al.*, 2022). In this case, emissions reduction shifts the whole distribution downward, lowering the median and the 95th percentile by similar amounts, but does not narrow the upper tail, whose spread above the median (0.99 and 0.98 m at 1.8 and 3.6 °C) is set by WAIS. The probability of exceeding 2 m relative to pre-industrial is 3% under the 1.8 °C trajectory and 6% under 3.6 °C. The 3% lower bound on this range highlights the fact that emission reductions alone cannot retire the high risk outcomes, though it is predicated on the values in the AR6 low-confidence scenarios that arise more from structured expert judgment than from data and reliable physical models. Emerging research on processes absent from current ice-sheet models points toward mechanisms that could raise those values (Section 2.2.2).

The risks and costs to coastal communities helps to highlight the dichotomy between the two possible scenarios of fast vs slow WAIS mass loss, or equivalently, emissions-insensitive vs emissions sensitive hazards (Figs. 10 and 11a). The difference at the 5% exceedance probability, one possible design threshold for coastal adaptation (*van de Wal et al.*, 2022) (Fig. 10b), is 0.91 m of sea-level rise by 2100 under the current policy warming trajectory (2.7 °C). This difference translates into roughly 170M more people exposed to annual flood risks, US$122B/yr in coastal adaptation, and US$12.2T/yr (about 10% of global GDP per year) in coastal flood damages without further adaptation. The damages are interpolated linearly between the values reported by *Jevrejeva et al.* (2018). For intuition, the fast-WAIS scenario at 5% exceedance implies more than seven times more sea-level rise in the next 75 years than was experienced in the past 125 years, while the slow-WAIS scenario implies more than three times the sea-level rise.

Resolving the question of WAIS mass loss rates is among the most financially consequential goals in modern science (Fig. 11c). A value-of-information analysis shows that knowing whether WAIS will lose mass slowly or rapidly is worth roughly US$4.6 trillion in avoided costs. Those costs combine adaptation costs (*Tiggeloven et al.*, 2020) with flood damages from *Jevrejeva et al.* (2018), which we interpolate linearly between their reported values and extrapolate linearly beyond 1.8 m (Supplementary Material). That figure is the maximum over the range of prior confidence in slow WAIS mass loss, reached at 76% confidence for information arriving today. At the $p(\mathrm{S1}) = 90\%$ case adopted here the value is US$3.8T. The value decays with time because late-arriving information leaves fewer years for decisions to improve. These numbers frame the research investment question in concrete terms: even partial reduction of WAIS uncertainty, well short of perfect knowledge, would justify annual research expenditures orders of magnitude above current levels, $\sim$ $10M *Catania et al.* (2022). The return on that investment is determined by how quickly the observations can narrow the range of WAIS outcomes, which depends on both the rate of data collection and the development of actionable process-based model results, like those used in the IPCC, capable of providing explanatory insights that underscore deeper understanding of ice sheets and climate.

Scientific research is a necessary condition for providing actionable information for coastal adaptation, but it is as likely as not that further research will add to the growing evidence to support rapid WAIS mass loss rather than build confidence in a slow-WAIS future (*Fricker et al.*, 2025). Sustaining a wide range of outcomes that admit ≤ 0.74 m and ≥ 1.92 m of GMSL rise in 2100 at 5% probabilities, will force decisions on coastal planning, adaptation, and managed retreat that impact millions of lives and cost billions of dollars to be made

under large uncertainties. The extent of these uncertainties in terms of lives, property, and ecosystems raises the question: Can we take action to increase the potential stability of West Antarctica? How much harm to people and the environment can we prevent? Recently, some research directions have been proposed to answer this question (*MacAyeal et al.*, 2024) with initial models of glacier interventions aimed at stabilizing WAIS show promise (*Meyer et al.*, 2026; *Miller et al.*, 2026). In the context of this study, stabilizing WAIS means increasing p(S1) and reducing the 95% (5% exceedance) of S2, the scenario that admits rapid WAIS mass loss. As with any nascent research, there are many unresolved questions about the feasibility, unintended consequences, and governance of these interventions (*Siegert et al.*, 2025). There are risks in intervening to remediate the West Antarctic Ice Sheet and slow its loss. Similarly, the logistical challenges will undoubtedly be great, but these risks and challenges should be weighed against the risks and challenges of coastal adaptation for sea-level rise. There is no response to sea-level rise that comes with zero risk and zero cost.

5 Conclusions

We focus on projecting 21st-century sea-level rise with an intermediate-complexity model in which calibration, validation, and uncertainty quantification are primary features. SPECTRUM represents each contributor with a parsimonious physical model, including a new ocean-forced model of Greenland outlet-glacier discharge. Each is calibrated by Bayesian inference against independent observational records where such records exist, and emulated from regional climate models or adopted from published sensitivities where they do not. With observed GMSL withheld from every calibration, the summed components close 94–103% of the satellite-altimetry rate. They reproduce the level of multiple independent reconstructions within error since approximately 1950 and 1.2 °C of warming, and they reproduce the observed sensitivity of the rate of rise to warming and a 1950–2000 rate within the range of the observational records. Because the validation window is comparable in duration to the forecast horizon and in warming to the 2.7 °C scenario, this hindcast skill supports forecasting the calibrated components through 2100. The 3.6 °C forecast extends about twice as far beyond the validated warming. Where WAIS is assumed very likely to continue its observed mass-loss trajectory (p(S1) = 90%), SPECTRUM at 2.7 °C gives a median of 0.60 m, 23 mm and 0.13 m below the extrapolation of observed GMSL trends (0.63 m) and semi-empirical models (0.73 m). It projects medians of 0.46 and 0.78 m at 1.8 and 3.6 °C, 3 to 13% above AR6 medium confidence. Under an assumed weighting of 90% toward accelerated WAIS loss (p(S1) = 10%), the medians rise to 0.60, 0.75, and 0.94 m. The 90% credible interval at 2.7 °C widens from [0.47, 0.83] m to [0.54, 1.74] m.

Under the accelerated weighting, the uncertainty in these forecasts has two sources. The first is the set of contributors whose temperature sensitivity we calibrated with observations. These account for 1 to 6% of the forecast variance and 15 to 27% of the width of the 90% credible interval. Their contribution shifts systematically with warming, and emissions cuts reduce it. The second is the West Antarctic Ice Sheet, which accounts for more than 90% of the forecast variance and for 73 to 85% of the credible interval. WAIS responds weakly to the emissions pathway. Holding warming to 1.8 °C rather than 3.6 °C lowers the median by 0.34 m but moves the probability of exceeding 2 m above pre-industrial only marginally, from 6%

to 3%, a change that is small relative to the weak physical constraints on the tail. Beyond 2100 the picture changes, because 21st-century warming sets the long-term commitment from Antarctica (*Lowry et al.*, 2021; *Seroussi et al.*, 2024). Under the slow weighting, the calibrated contributors set 43 to 61% of the interval.

The two sources of uncertainty call for different responses. Emissions cuts reduce the relatively predictable, temperature sensitive part of sea-level rise, which includes the thermosteric, glacier, and Greenland Ice Sheet contributions. Resolving the rate of West Antarctic mass loss, which recent studies have found to be relatively insensitive to emission reductions this century, would narrow the range that coastal communities must prepare for. We estimate that information to be worth up to about US$4.6 trillion in avoided damages and adaptation costs, a value that declines the later the information arrives. The rate of West Antarctic ice loss over the next two decades is likely the most informative measurement, along with more detailed and persistent monitoring of key areas of West Antarctica to inform forecast models. The cost of such monitoring is paltry compared with the cost of not resolving the question of West Antarctic mass loss. However, emissions cuts and better data and models may not reduce the 21st-century tail risk of rapid sea-level rise, although emissions cuts do reduce the long-term commitment from West Antarctica (*Lowry et al.*, 2021). Whether interventions aimed at slowing West Antarctic ice loss could do so is an open question.

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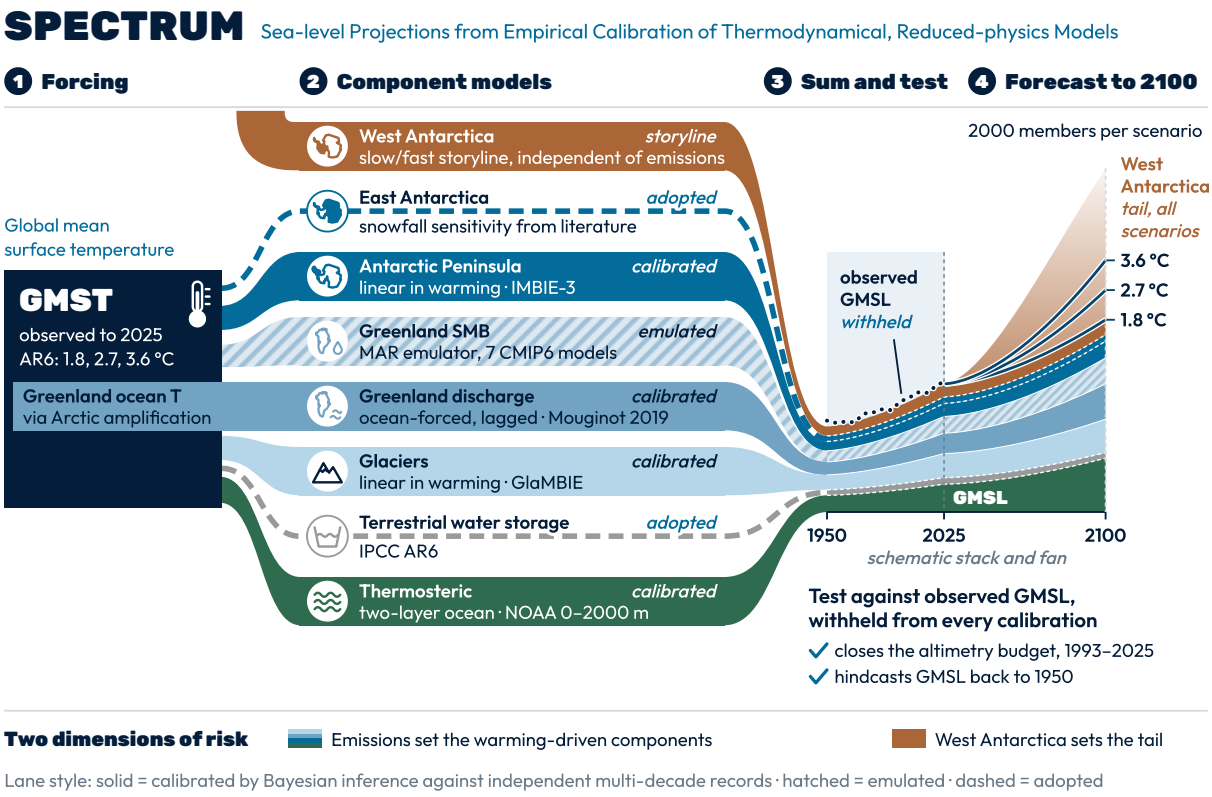


Figure 1: **The SPECTRUM framework.** (1) Global mean surface temperature (GMST), observed to 2025 and following the AR6 1.8, 2.7, and 3.6 °C paths, forces each component. Greenland discharge is forced by ocean temperature around Greenland, derived from GMST through Arctic amplification. (2) Each contributor has its own model. Line style marks how the model is constrained: solid lanes are calibrated by Bayesian inference against independent multi-decade records, the hatched lane is emulated, and dashed lanes are adopted from published sensitivities and assessments. West Antarctica (brown) follows a slow/fast storyline that is independent of emissions this century. (3) The components are summed and tested against observed GMSL, which is withheld from every calibration. (4) The forecast to 2100 uses 2000 members under each warming level. Emissions set the warming-driven components, and West Antarctica sets the tail in all scenarios. The stacked areas and fan are schematic.

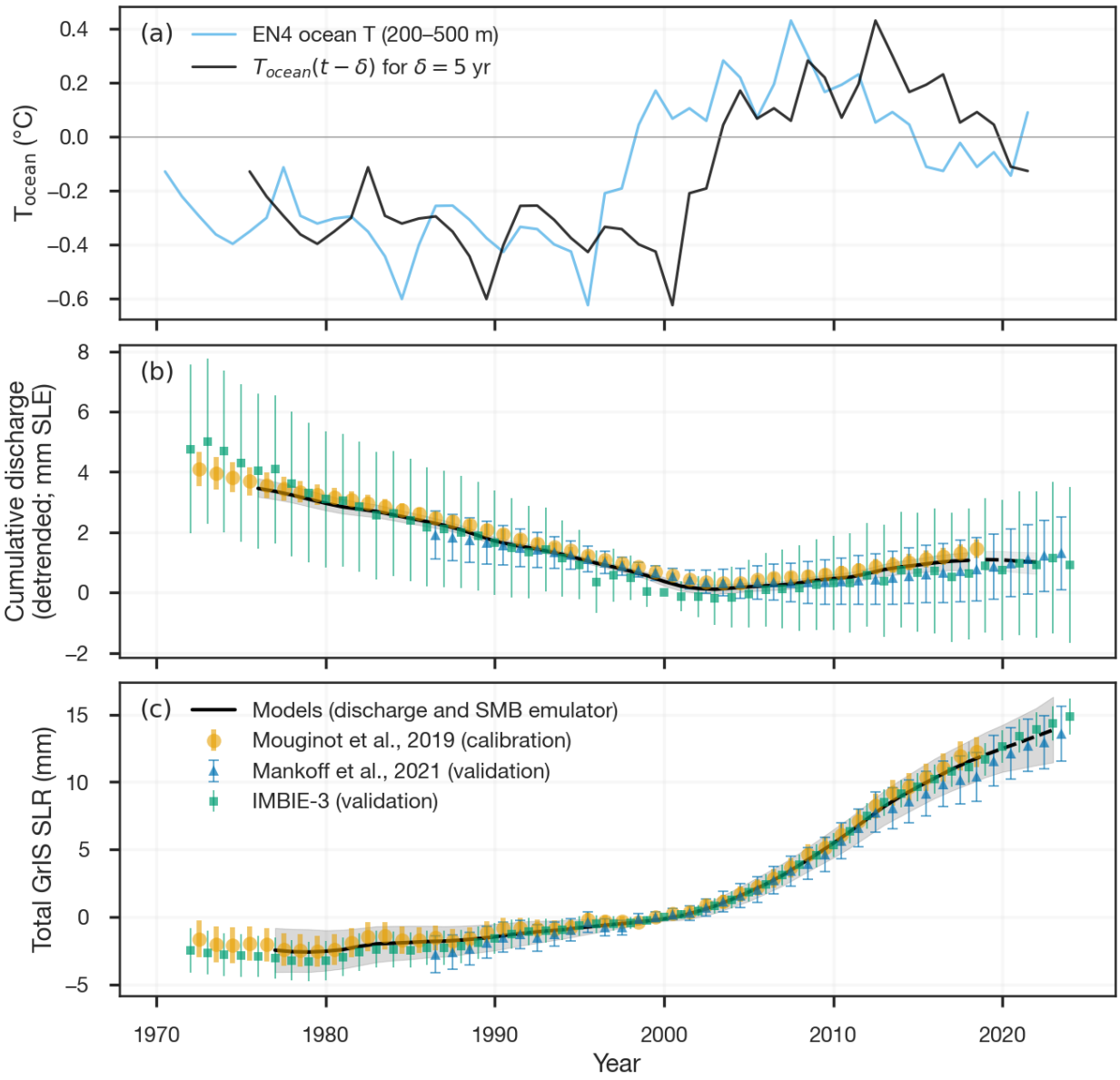


Figure 2: **Calibration and validation of the Greenland discharge model. (a)** Subsurface ocean temperature anomaly relative to 1995–2005 (from EN4, averaged over depths of 200–500 m and over 58–80°N, 75–5°W, with annual resolution; blue) and the same signal delayed by $\delta = 5$ yr, $T_{\text{ocean}}(t-\delta)$ (black). **(b)** Detrended cumulative Greenland ice discharge, showing the model fit (black) with its 90% credible interval, compared with the calibration data (*Mouginot et al.*, 2019) (orange) and the time series of *Mankoff et al.* (2021) (blue) and IMBIE-3 (*Otosaka et al.*, 2026) (green) used for validation. Each record is detrended by the background rate r_0 from a fit of the same model to that record. **(c)** Total Greenland contribution to GMSL, the sum of the modeled SMB and discharge contributions (black, with 90% credible interval). SMB comes from a statistical emulator of the MAR regional climate model. Dashed lines continue the models past the end of the *Mouginot et al.* (2019) record.

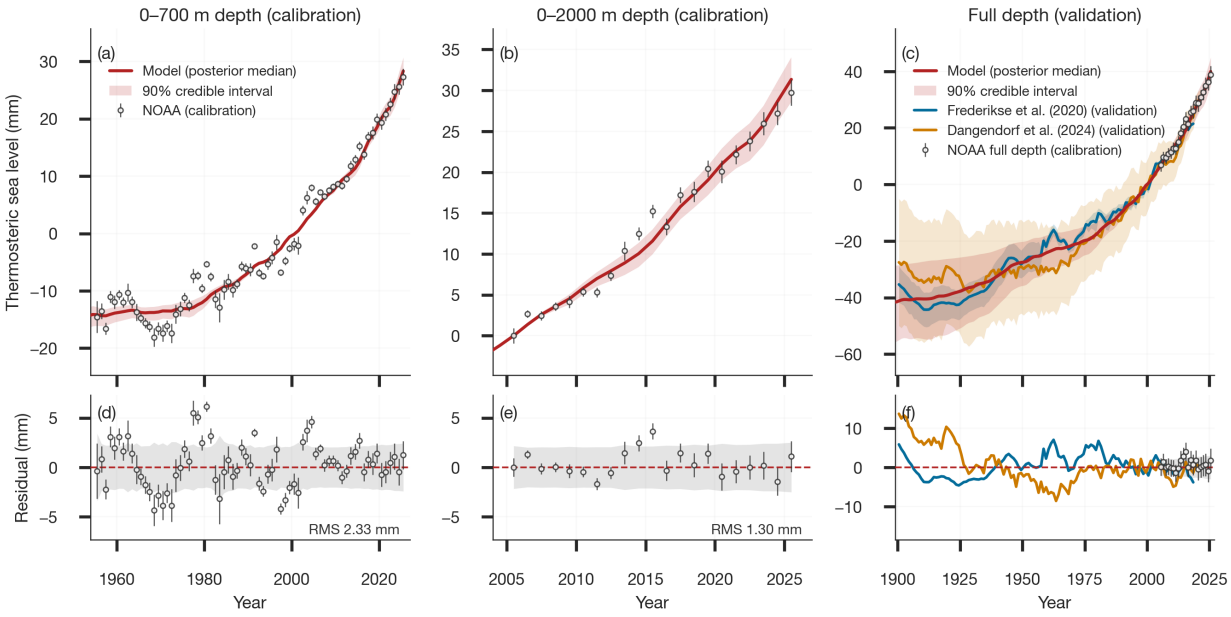


Figure 3: **Two-layer thermosteric model fit and validation.** (a) Posterior median of the upper layer model H_{u} (Eq. 8) and 90% credible interval fitted to observations at 0–700 m depth (circles with error bars; from NOAA (*Levitus et al.*, 2012)). (b) Same as in (a) but for the 0–2000 m data and H_{ud} (Eq. 11). (c) Validation of the full-depth model H_{full} (Eq. 12) against the results from *Frederikse et al.* (2020) (blue) and *Dangendorf et al.* (2024) (orange). Also shown for reference are the 0–2000 m observations (circles, same as in panel b). (d–f) Residuals corresponding to (a)–(c), with the gray band showing the 90% credible interval on the residual and the RMS residual reported in each calibration panel.

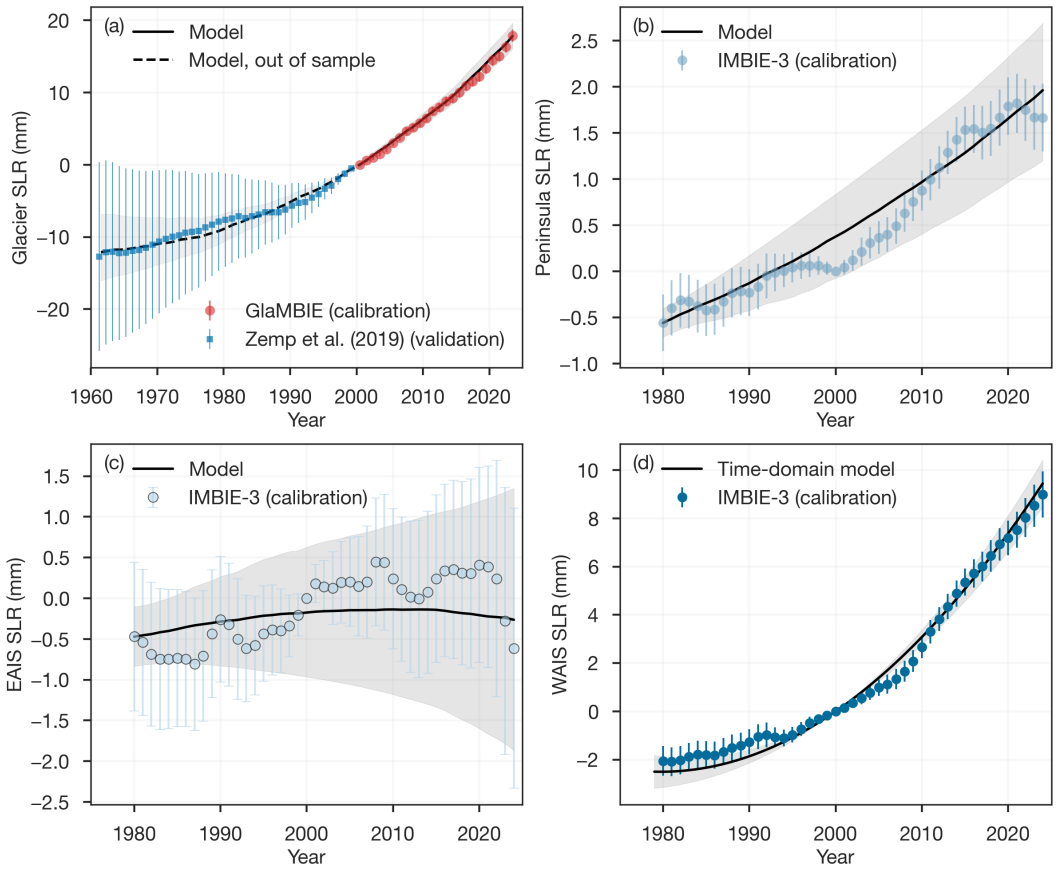


Figure 4: **Per-component calibration fits. (a)** Glaciers, **(b)** Antarctic Peninsula, and **(c)** EAIS observed cumulative contributions to GMSL (circles with 90% error bars) are shown with the respective posterior median models (black lines) and their 90% credible intervals (gray shading). **(d)** The WAIS cumulative contribution is shown with a quadratic fit in time rather than in temperature (black line) and its 90% credible interval (gray shading). In (a) the glacier model is also run back to 1961 (dashed line and lighter shading, outside the calibration period) and compared with the validation record of *Zemp et al.* (2019) (squares), whose cumulative values are matched to the model at the end of 1999. Data are from *The GlaMBIE Team et al.* (2025) (calibration) and *Zemp et al.* (2019) (validation) for glaciers and *Otosaka et al.* (2026) for the ice sheets.

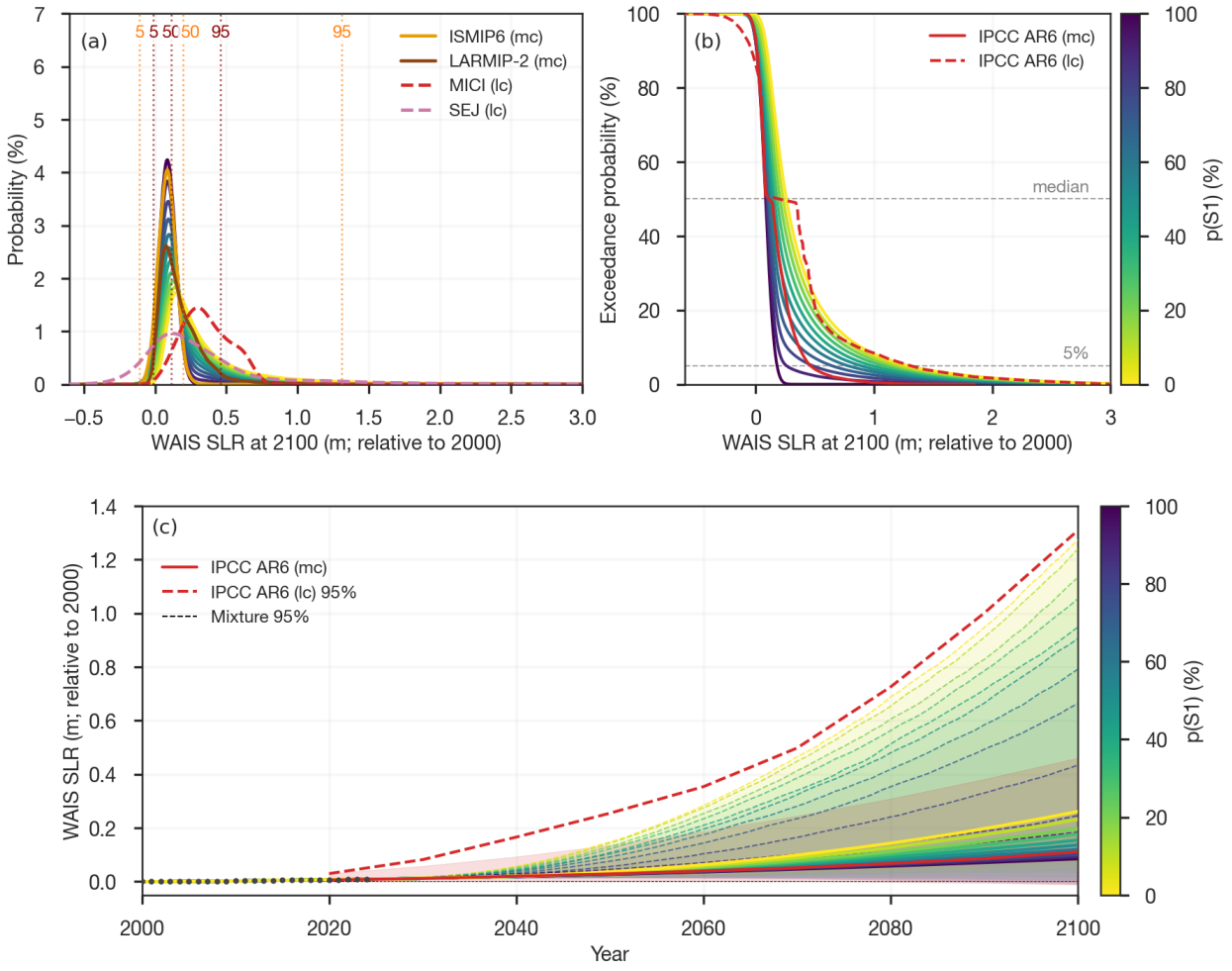


Figure 5: **WAIS scenario mixture model. (a)** Probability density of the WAIS contribution to GMSL at 2100 (relative to 2000) for an 11-member sweep of the mixture weight p(S1) from 100% (S1, continued observed trend) to 0% (S2, accelerated loss), colored by weight (colorbar). Also shown are the IPCC AR6 model ensembles for the whole Antarctic Ice Sheet that underlie its medium-confidence projections at 2.7 °C (mc; ISMIP6 emulator, *Edwards et al.*, 2021; LARMIP-2 ensemble, *Levermann et al.*, 2020) and its low-confidence projections at 4.4 °C (lc; marine ice cliff instability, MICI, *DeConto et al.*, 2021; structured expert judgment, SEJ, *Bamber et al.*, 2019). Dotted vertical lines mark the 5th, 50th, and 95th percentiles of the AR6 medium-confidence (red) and low-confidence (orange) projections. **(b)** Corresponding exceedance probability curves for the same p(S1) sweep, with the IPCC AR6 medium-confidence 2.7 °C (red solid line) and low-confidence 4.4 °C (red dashed line) exceedance curves for the Antarctic Ice Sheet overlaid for reference. **(c)** Time series for the p(S1) sweep and the IPCC AR6 medium-confidence 2.7 °C projection (red solid line with 5–95% range shaded) and low-confidence 4.4 °C 95th percentile (red dashed line). The thin dashed lines show the 95th percentile for the correspondingly colored p(S1).

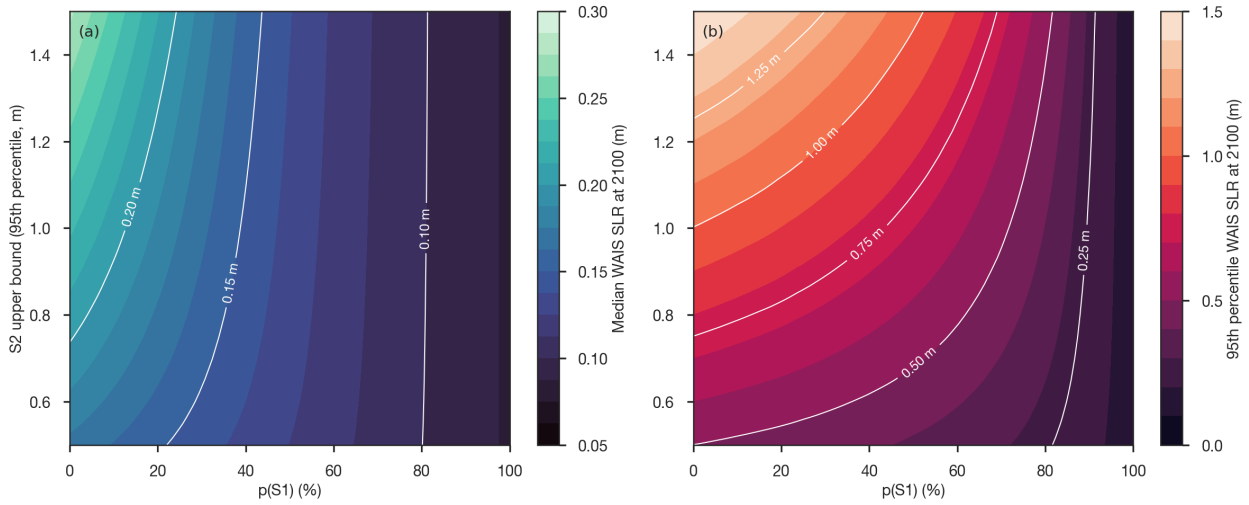


Figure 6: **WAIS contribution to GMSL at 2100 as a function of the two mixture parameters. (a)** Median and **(b)** 95th percentile of the S1/S2 mixture (relative to 2000) across the S1 probability, $p(\text{S1})$, and the S2 95th percentile at 2100.

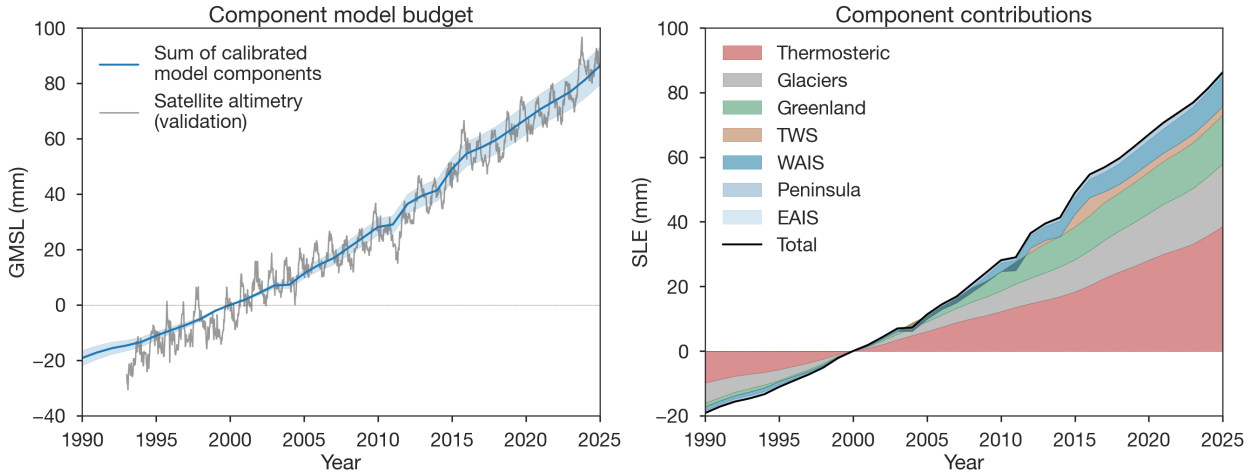


Figure 7: **Budget closure (validation): sum of model components versus observed GMSL. (a)** Sum of components (blue, with shading for its 90% credible interval) and NASA satellite altimetry (gray) (*Beckley et al.*, 2017; *Hamlington et al.*, 2020). **(b)** Stacked per-component contributions and their sum in black. Terrestrial water storage enters only from 2003, the start of the GRACE record from which it is derived, and is held at its 2018 value thereafter.

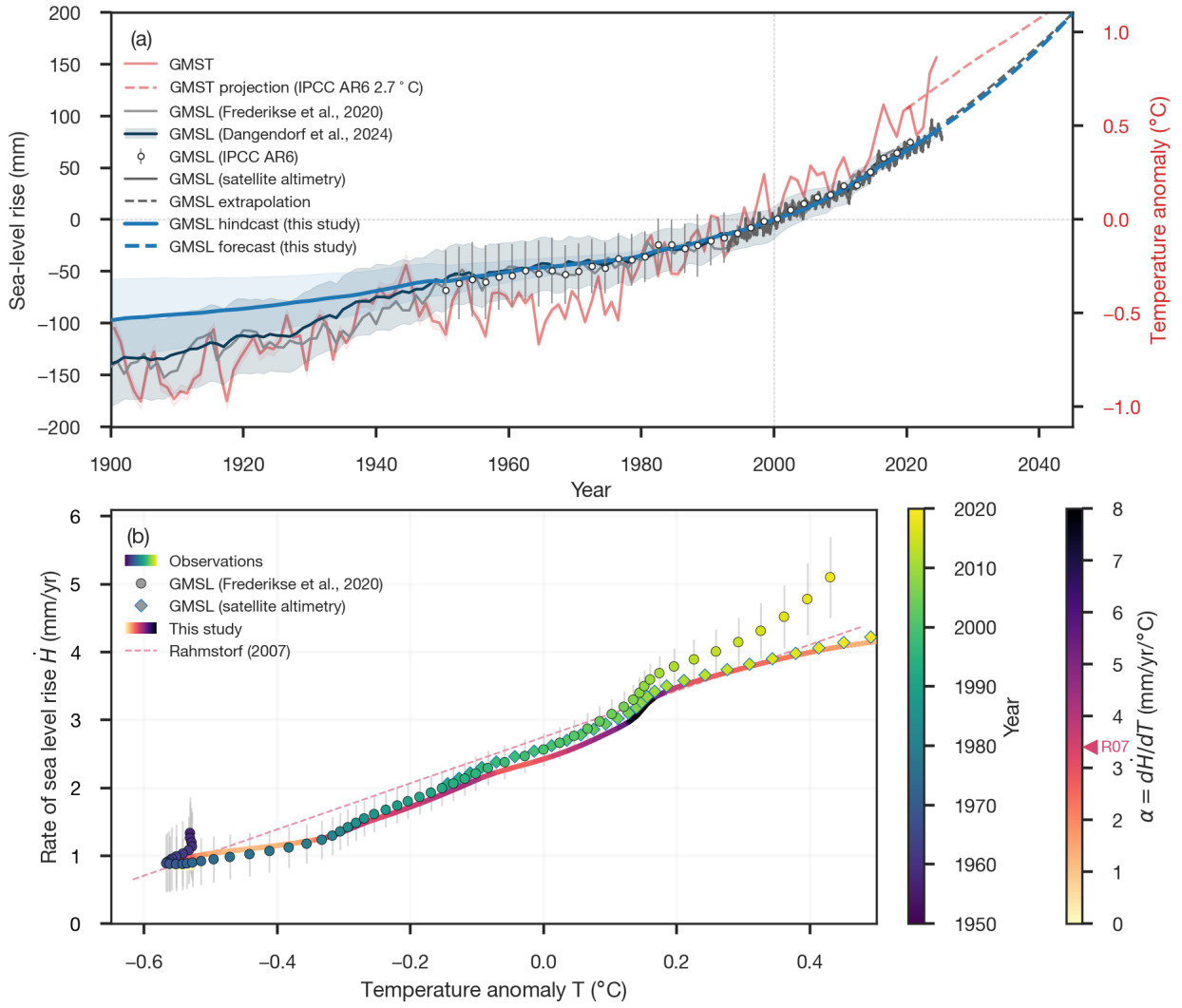


Figure 8: **Global mean sea-level rise and global mean surface temperature. (a)** GMSL reconstructions from *Frederikse et al.* (2020) (light gray) and *Dangendorf et al.* (2024) (navy, with 90% uncertainty range shaded), the IPCC AR6 observational compilation (*Fox-Kemper et al.*, 2021) (white circles with 5–95% range bars), and NASA satellite altimetry (*Beckley et al.*, 2017) (mid gray, with the satellite-era quadratic of *Hamlington et al.* (2024) extrapolated as a dashed line) show sea level rising with warming. The hindcast of this study is shown in solid blue with shading for its 90% credible interval, and the median forecast for the 2.7 °C scenario in dashed blue. Before 2000 the hindcast includes the prescribed contribution of uncharted glaciers (*Parkes and Marzeion*, 2018). GMST (red) (*Rohde and Hausfather*, 2020) is referenced to the right-hand axis. **(b)** Rate of GMSL rise ($\dot{H} = dH/dt$) versus GMST anomaly, T. Observed values are shown in circles (*Frederikse et al.*, 2020), colored by year, with 90% error bars that account for the correlated errors of a cumulative record, and diamonds show the rate of the satellite-era quadratic of *Hamlington et al.* (2024). The observed rates include terrestrial water storage, which the model sum sets to zero before 2003. The component model of this study, the median of the summed component ensemble without blending, is a line colored by instantaneous transient sensitivity $\alpha = d\dot{H}/dT$, with rates from the same 30-yr local regression as the observations.

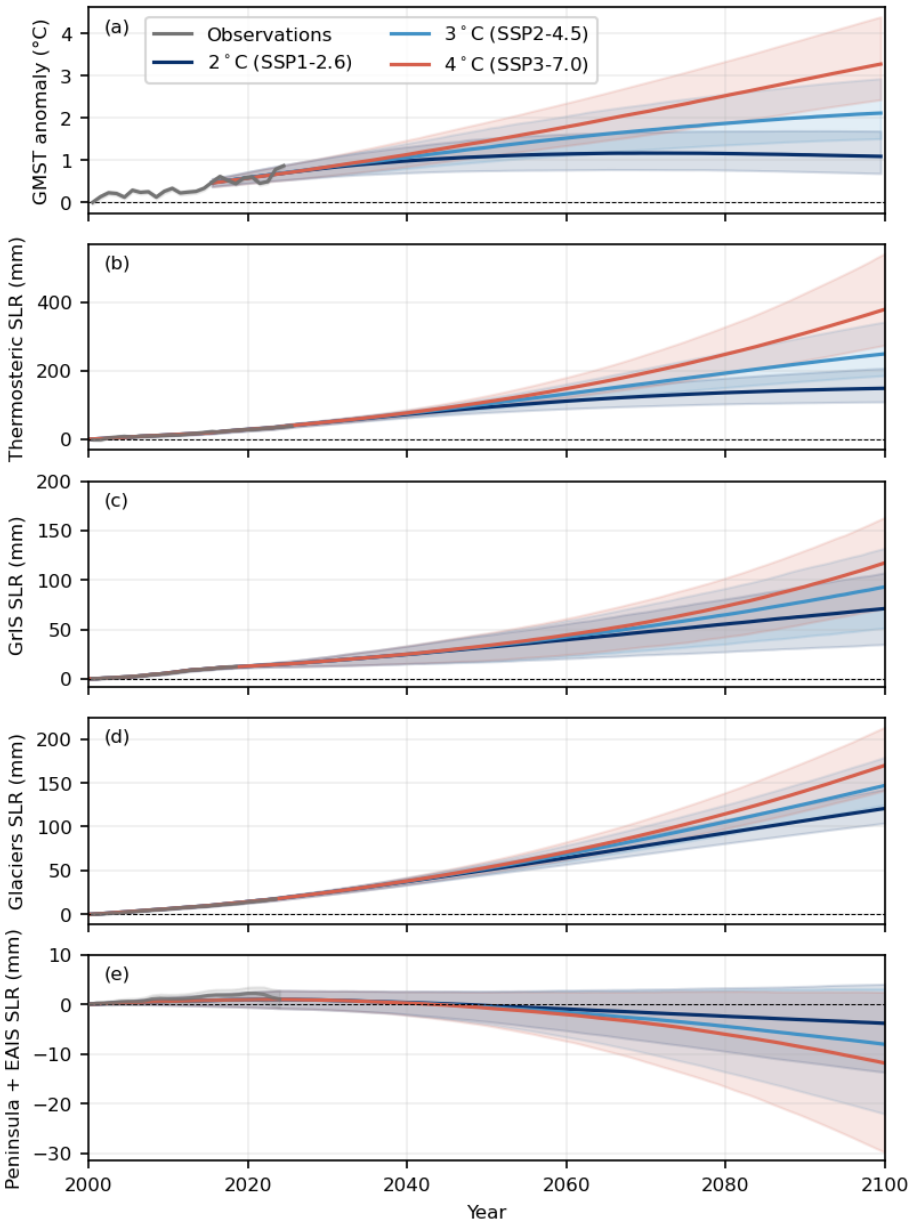


Figure 9: **Component-level forecasts under three warming scenarios.** (a) Observed global mean surface temperature (GMST) anomaly (gray) relative to 2000, with the IPCC AR6 GMST trajectories for the 1.8, 2.7, and 3.6 °C scenarios and their 5–95% ranges (shaded). The curves show warming relative to 2000. The legend labels each scenario by its AR6 central-estimate warming over 2081–2100 relative to the pre-industrial (1850–1900) mean. (b) Thermosteric sea-level contribution relative to 2000, with observations (gray) and the forecast median with its 90% credible interval from the full-depth two-layer model driven by the corresponding GMST trajectory. (c) Greenland Ice Sheet, (d) glaciers, and (e) the sum of the Antarctic Peninsula and East Antarctic Ice Sheet (EAIS) contributions, each shown as in (b).

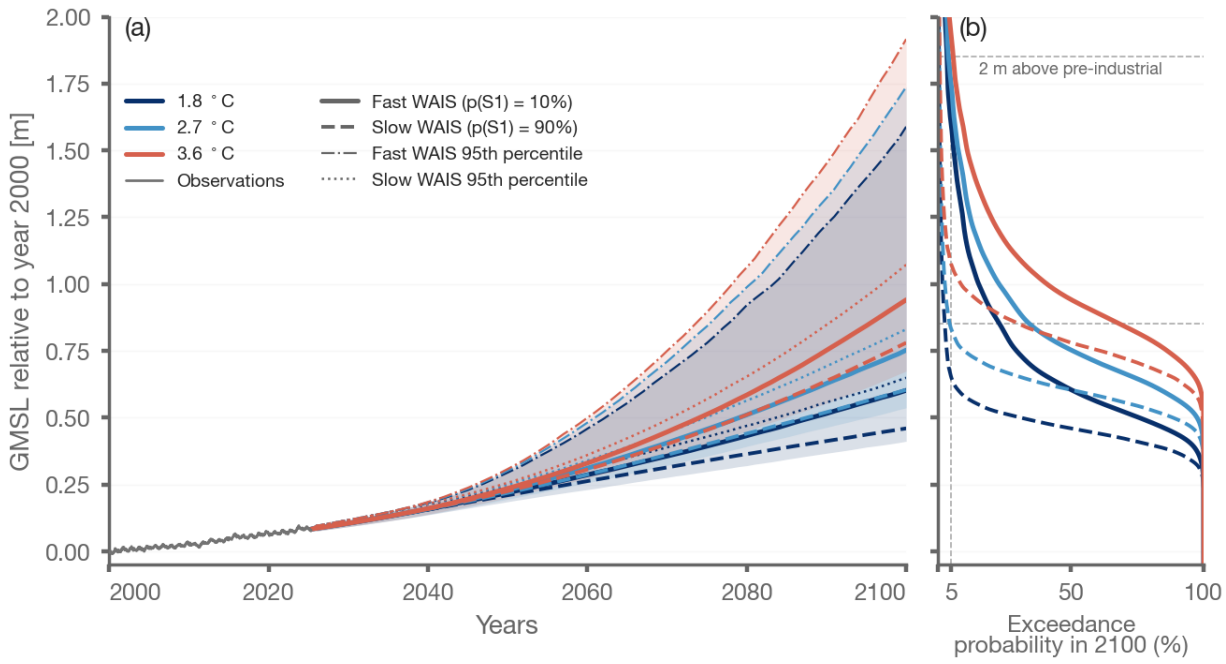


Figure 10: **Global mean sea-level projections.** (a) Global mean sea level relative to 2000 under 1.8, 2.7, and 3.6 °C of warming, with satellite altimetry observations (gray) (*Beckley et al.*, 2017). Solid lines show the median of the fast-WAIS storyline (p(S1) = 10%) with its 90% credible interval (5th–95th percentiles, shaded). Dashed lines show the median of the slow-WAIS storyline (p(S1) = 90%), whose interval is not shaded. Dash-dot and dotted lines mark the 95th percentiles of the fast-WAIS and slow-WAIS forecasts, respectively. (b) Probability of exceeding each sea level in 2100 (horizontal axis) for the fast-WAIS (solid) and slow-WAIS (dashed) forecasts. The vertical axis is shared with (a), so sea level is relative to 2000. Thin gray dashed lines mark 1 m and 2 m above pre-industrial (horizontal) and 5% exceedance (vertical). The 95th-percentile lines in (a) reach the 5% exceedance level in 2100.

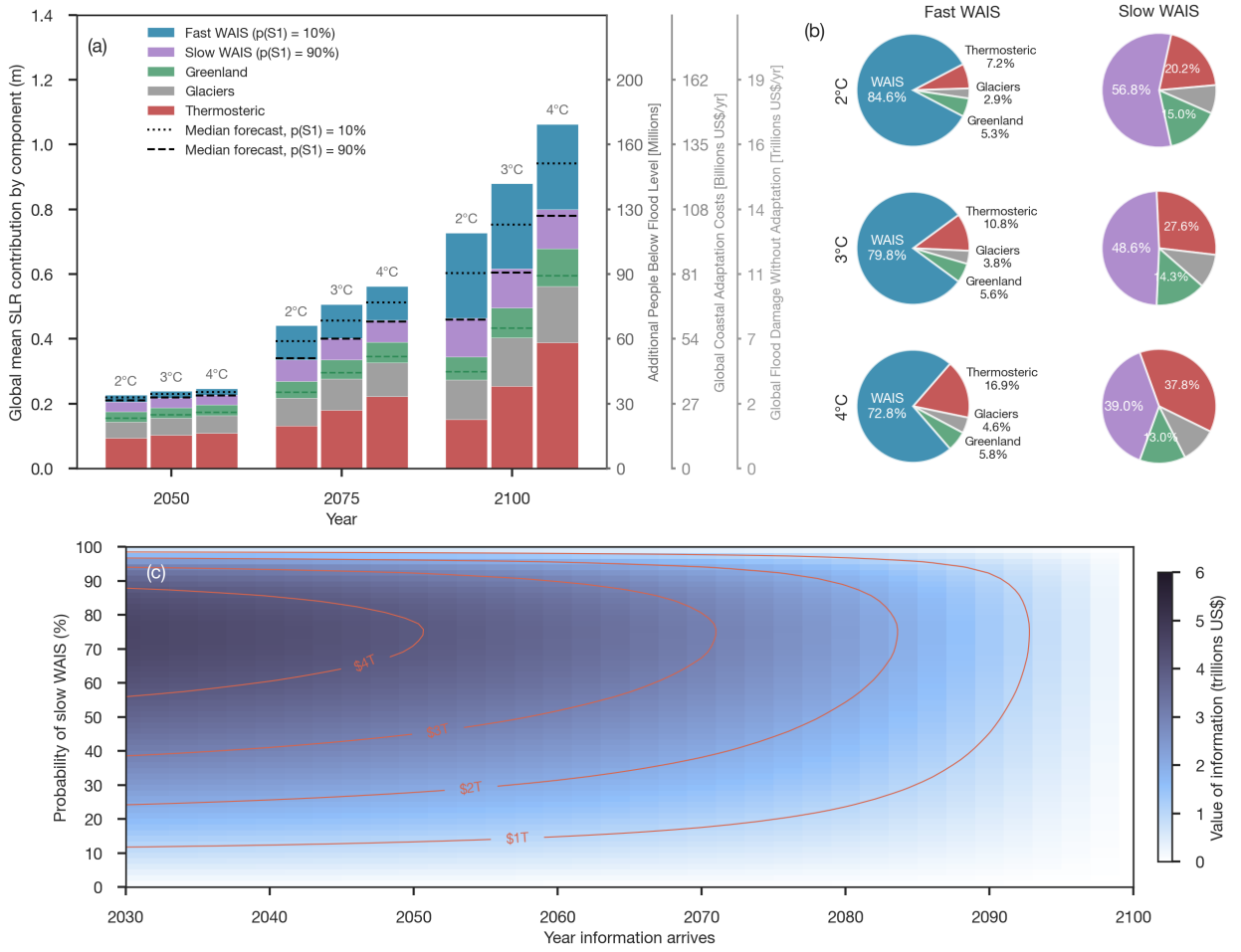


Figure 11: **Component contributions, variance decomposition, and value of information.** **(a)** 21st-century mean contribution to GMSL rise by component and respective societal costs. Green dashed lines within Greenland bars partition discharge anomaly (below) from SMB anomaly (above). The purple box shows the WAIS contribution under the slow-WAIS storyline, while the blue bar shows the additional potential SLR from rapid ice loss. The right-hand axes show previously published estimates of the additional number of people on land below flood level (*Kulp and Strauss*, 2019), global coastal adaptation costs (*Tiggeloven et al.*, 2020), and global flood damages without adaptation (*Jevrejeva et al.*, 2018), which we interpolate linearly between their reported values and extrapolate linearly beyond 1.8 m (Supplementary Material). **(b)** Shares of the 90% credible interval at 2100, taken as each component's own interval width as a fraction of the summed widths. **(c)** Estimated financial value of resolving WAIS uncertainty as a function of when information arrives. Contours and colormap in trillions US$ based on coastal adaptation costs (Supplementary Material). The plot shows that information has more value if it arrives early enough to inform key decisions and if it provides sufficient confidence that WAIS will lose mass slowly, allowing planning around lower future sea levels.